C2S2 ”Kestrel” SAR ADC Theory of Operation and Architecture Comparative Analysis

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Active Work in Progress

Absolutely no AI was used in writing this.

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1 Overview

This document will outline the detailed theory of operation of the FA24SP25 semester fully and truly differential 8-bit successive approximation register analog-to-digital converter (SAR ADC), codenamed Kestrel. This document will try and do so mostly from the ground up; circuits, digital logic, and basic FET, knowledge is assumed and ADC/DAC specific knowledge + basic signals knowledge will ease the reading. Furthermore, a comparative analysis is incorporated, rationalizing the C2S2 SAR ADC design decisions.

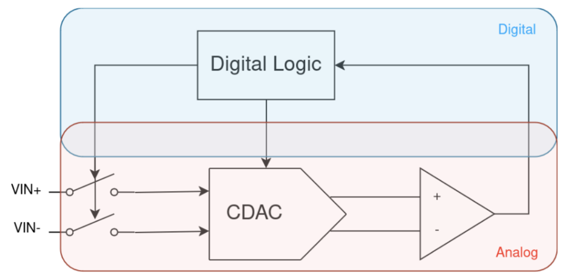


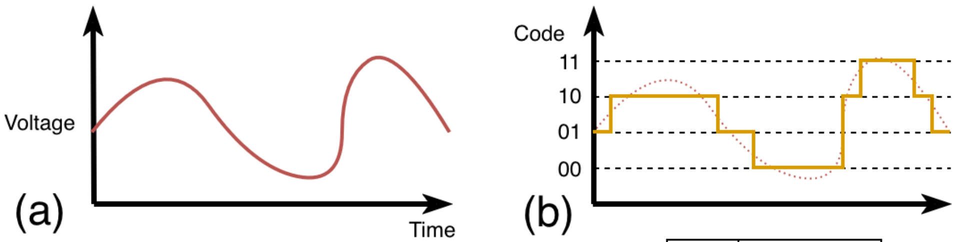
Figure 1: High-level Block Diagram

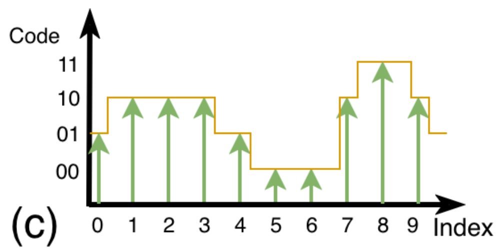
2 General SAR ADC introduction

2.a ADC Introduction and Topologies

An Analog-to-Digital converter (ADC) is a type of data converter that converts continuous-time and continuous-voltage (Analog) signals such as audio, temperature, and motion into discrete-time quantized digital values to be used for further processing (e.g. via digital signal processor, software, FFT...) or digital storage (e.g. SRAM, DRAM, Flash memory, CD-ROM, etc.).

Two processes occur simultaneously within an ADC: sampling and quantization. Quantization is the process that divides a continuous quantity into one that can only take on those from a finite set of values. Sampling is the process of turning a signal that is defined at all points in time to one that is defined only at certain integer indices. Simply put, this turns a "smooth" signal that can take on infinitely many values at all times into a "stepped" signal that can take on only finitely many certain values at certain points in time. An inverse called a Digital-to-Analog converter (DAC) exists as well, which converts digital values into analog voltages.





(d)

i	Data
0	01
1	10
2	10
3	10
4	01
5	00
6	00
7	10
8	11
9	10

Figure 2: (a) Analog signal (b) Analog signal undergoing quantization (c) Quantized signal undergoing sampling (d) Digital signal

Many variants of ADCs exist and can be categorized in a variety of different ways. One way to roughly categorize ADCs is in their mechanism in determining the digital value that corresponds to an analog voltage [1].

a) **Parallel Search:** These type of ADC determine the digital value in a single step. An example of such an ADC is the Flash ADC, which uses $2^n - 1$ comparators to determine which digital value corresponds to the analog input. This type of ADC is often used in lower-resolution, high-speed applications. Many variants of the Flash ADC exist, such as folding and interpolating ADCs. More information can be found in [1].

b) **Sequential Search:** These types of ADCs determine the digital value successively by comparing multiple times, consecutively. An example of such an ADC is the SAR ADC, which will be discussed in greater detail below. This type of ADC is often used in moderate-resolution, moderate-speed applications. There are others such as pipeline ADCs are in this category as well.

c) **Linear Search:** These types of ADCs determine the digital value in a linear search, such as simply measuring the time it takes to charge a capacitor until the capacitor voltage reaches the input voltage (single-slope converter). Variants of these (dual-slope and multislope) architectures are fascinating and blur the lines between these types of data converters. These ADCs have been present in some high precision multimeters such as the HP/Agilent/Keysight 3458A [2].

d) **Oversampled:** Such ADCs include $\Delta\Sigma$ ADCs. These types of ADCs are *not nyquist-rate* converters; this means that these ADCs sample at a rate much higher than the Nyquist frequency (2x the maximum input frequency), unlike the other converters mentioned previously. These data converters are typically low-speed and high-resolution. Although this seems counter-intuitive, a hand-wavey explanation for this is that these converters leverage digital signal processing to trade off their high sampling rate for resolution, enabling ENOBs excess of 20 bits. This ADC is not covered in detail here, and this short excerpt definitely does not do this architecture justice.

Modern state-of-the-art ADCs do not fall in a single category and often blur the lines or combine multiple architectures together. For example, Noise-Shaping SAR ADCs combine both oversampled and sequential search techniques [3]. Interleaved ADCs do not clearly fall in any of these categories either.

2.b SAR ADC General Operation

SAR (Successive Approximation Register) ADCs use the sequential search paradigm. A SAR ADC internally has an analog comparator, a DAC, digital control logic, and an input sampling mechanism. A rough diagram of this can be found in 1. Note that there is no explicit input sampling mechanism block as the input sampling is done by the CDAC and the switches left of the CDAC.

A good physical analogy for the principle of operation of SAR ADCs is a balance scale with reference weights. When weighing an object, an object of unknown weight is placed on one side of the scale, and reference weights are successively added and removed to the other side of the scale depending on the direction the scale tips. This is repeated until a weight can be estimated with the reference weights. A person familiar with algorithms may notice this is similar to a binary search where multiple greater than or less than comparisons are done sequentially to find a the index of a value in a sorted list. This is a useful analogy in some cases; however, a SAR ADC need not use a binary search, some other search algorithms (which are more power efficient) can be used instead.

A SAR ADC samples, or ”grabs” the current voltage at, the input of the ADC at that moment in time with the input sampling mechanism (similar to the act of placing unknown weight on one side of the scale), guesses with a known voltage with the DAC (similar to the act of placing a reference weight on the scale), and uses the comparator to determine whether the voltage outputted by the DAC is higher or lower than the sampled input voltage. The output of the comparator informs an algorithm to generate a closer voltage via the DAC. This repeatedly happens, and the DAC is able to generate voltages that are increasingly close to the input of the ADC.

There are many components in the design of a SAR ADC including the algorithm that generates DAC voltages, the input sampling mechanism, the comparator, and the DAC itself. These considerations are critical to the SAR ADC’s performance in terms of power, resolution, and speed.

2.c ADC Performance Metrics

An ADC designer or end user often wants to characterize the performance of the ADC to determine whether an ADC fits a required design specification. Some of these metrics are listed below:

a) **Input Voltage Range:** This is the acceptable range of voltages that the ADC can accept. In a simple, single-ended ADC with a single input (not in the case of the FA24SP25 ADC), this would be the minimum and maximum input voltage that the ADC can accept. Any voltage outside of this range may, in the best case, not be resolved properly and/or clipped, or in the worst case, permanently damage the ADC. This is somewhat more complicated in differential ADCs (two input ADCs) and will be discussed in greater detail later.

b) **Resolution**: This is one of the most important aspects of an ADC. Roughly speaking, this is how precisely an ADC turns an analog signal into a digital quantized level. An n-bit ADC will be able to turn a voltage (limited to a specified input voltage range) to 2^n digital levels. For example, a simple single input (single-ended) 8-bit ADC could output values ranging from 0 to 255. If the input voltage range of this ADC is 0V-1V, each increase of the input voltage by $\frac{1V-0V}{2^8} \approx 3.9mV$ will correspond to an increase of the digital output (code) by 1. Some ADCs claim to have resolutions of excess of 32 bits. At first glance, they will appear to output 32 or 64 bits; however, a 32+ bit ADC would have to be able to resolve less than a nanovolt for if it were to have a reasonable input voltage range (1-10V). This is typically lower than the thermal noise floor for most components at room temperature! Thus, many of the least significant bits are meaningless because they are usually just noise (the ADC cannot actually resolve them). This emphasizes the importance of ENOB, or *effective number of bits*. This describes the number of *useful* bits output by the ADC. Although an ADC may claim to be a 32 bit ADC, its ENOB may be much worse at 12 or 16 bits. [1]

c) **Power:** Like all electronics, ADCs require a certain amount of power to do their job. A lower power ADC results in one that can be used for longer when powered by battery, for example.

d) **Speed/Bandwidth:** This is typically represented by the sample rate of the ADC. This describes the number of times that the ADC converts an analog value to a digital value per second (quantization operation). For example, a 4MS/s (4 Megasample/Second) ADC will convert a continuous-time continuous-voltage analog signal to a discrete-time discrete-voltage digital value 4 million times per second. Therefore, the ADC will output a digital value 4 million times per second. The data rate that the digital logic receiving the ADC data has to handle will be effectively the sample rate of the ADC times the resolution that the ADC outputs. An exception to this is oversampling ADCs ($\Delta\Sigma$ ADCs), which need a more intermediate signal processing and will not be discussed in detail here.

e) **Die Area:** This represents the physical amount of area that the design takes up. Flash ADCs

[1] There are numerous metrics that contribute to the calculation of ENOB, making this a complicated topic very quickly. Some metrics include DNL (Differential NonLinearity), INL (Integral NonLinearity), SFDR (Spurious Free Dynamic Range). ENOB may also change depending on input frequency.

typically take up more area (for higher resolutions) than SAR ADCs for example. This is important in industry as it contributes greatly to the cost of the final product.

Figure of merits (FoM) are numbers representing a combination of the aforementioned characteristics. They allow engineers to quickly compare the "goodness" of ADCs. However, these are often imperfect because of the wide range of ADC applications (i.e. battery powered electronics may value power consumption over resolution and RF electronics may value bandwidth over power consumption). One FoM commonly used is the Walden FoM which is given by the following equation [4]:

$$FOM_W = \frac{P}{2^{ENOB} \cdot F_S} \tag{1}$$

Where FOM_W is the Walden Figure of Merit in (Joules/conversion step), P is the power in Watts, and F_S is the sampling frequency is samples/second. A lower FoM corresponds to a better ADC. Some other FoMs include the Schreier FoM.

3 Architectural Decisions and Comparison

3.a C2S2 Performance Targets

a) **Resolution: 8-bit ($\sim$ 8 bit ENOB)**. We believe this is a reasonable value for audio applications with careful design (bird call sampling) which should be achievable by C2S2.

b) **Sample Rate: $\sim$ 4Ms/S** This decision was made retroactively after solidifying the decision to use synchronous sampling.

c) **Power: As low as possible** We hope to target battery powered operation for bird tags.

d) **FoM: 0.1uJ/Conv** This seemed to be a reasonable target for SAR ADCs designed by C2S2 when comparing to state of the art SAR ADC FoMs.

e) **Die Area Usage: Less than** $1.5mm^2$ This is the tapeout area that we were allotted. The final die area usage was much smaller than this.

3.b DAC Architecture

We opt to use a CDAC (Capacitive Digital to Analog Converter) architecture within the SAR ADC. This implementation is popular among SAR ADCs. Compared to resistive DACs, CDACs do not have static power draw and thus are low power [5]. There are other DAC options such as current weighted DACs, but we opt to the CDAC because it is common.

3.b.1 Binary Weighted vs Split CDAC

Many CDAC topologies exist which tradeoff between accuracy, area, and power consumption. An important question to ask is whether to add a bridge capacitor or not, creating a split capacitor CDAC; this topic will be discussed further below[2].

In a binary-weighted (”original”) CDAC configuration, each bit is assigned a capacitance, which is scaled depending on the significance of the bit. A simplified CDAC design will be is shown below in Figure 3(a) below for the purposes of demonstration:

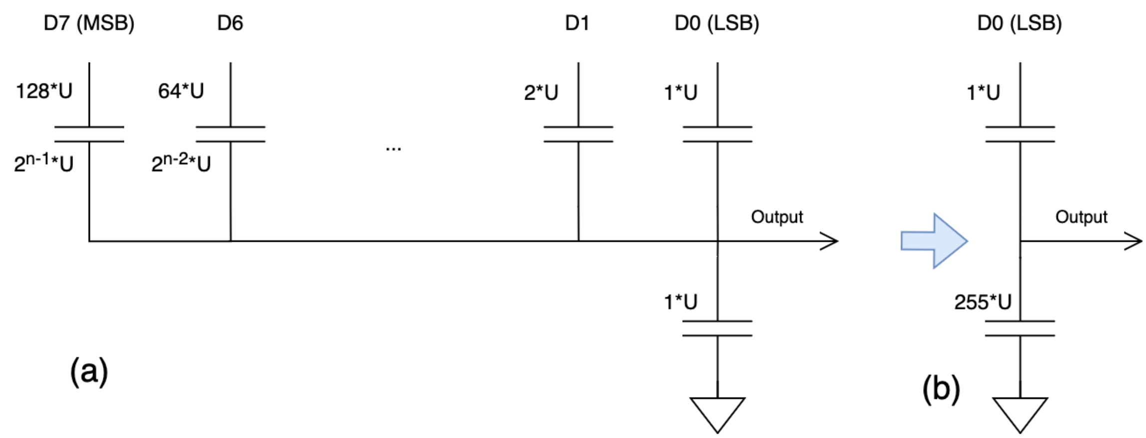


Figure 3: (a) Simple Binary-Weighted CDAC. (b) Equivalent representation where D1-D7 are low.

The LSB here is assigned a capacitor of $1U$ where ”U” is the ”unit” capacitance, or smallest value of capacitance present in the particular CDAC. The MSB for an 8 bit CDAC may be assigned $2^{8-1} = 128U$[3]. The operation of the CDAC can be explained via a practical example. Assume that the logic levels of the digital inputs (D0-D7) are 0V and 1V and the ”unit” capacitor is 1fF and all the capacitors are discharged at the start (all voltages at all points are 0V). If the LSB input (D0) is switched high, representing a binary input value of 8’b1, the 1U rightmost capacitor will have 1V present on the input plate of the capacitor. This will effectively form a capacitive divider represented in Figure 3b. Thus, the output will voltage will be $1V \cdot \frac{1U}{255U+1U} = 1/256V$. This capacitor divider can be rewritten for all digital inputs to find that counting up in binary will result in voltage increase in steps of $1/256V$. This can be generalized to the following equation:

$$V_{out} = V_{in} \cdot \frac{k}{2^n} \tag{2}$$

where k is the input value/code in binary, n is the number of bits in the DAC, and V_{in} is the logic level voltage input. This is a typical relationship for most DACs (not only CDACs). Note that for this 8 bit CDAC, the maximum output is 255/256, not 1. Furthermore, note that the output voltage

[2]Split capacitor may also be used to refer to a low power CDAC topology/switching algorithm, adding to the confusion in this topic. See Footnote 2 on Pg. 740 in: [6]

[3]An 8 bit SAR ADC does not necessarily have to have an 8 bit CDAC, as we will explore later

is proportional to the input logic voltage; thus, what is actually driving the CDAC (driver design) is important.

For CDACs with more than 8 bits of resolution, it may become cumbersome to add more capacitors – the number of capacitors required increase exponentially. The number of capacitors for an n bit binary weighted CDAC is equal to 2^n. To combat this issue, designers may opt to add a ”bridge” or ”attenuation” capacitor. In literature, there are two ways in which bridge capacitors are implemented: ones with a ”dummy” capacitor, and ones without [5, 7]. These two are not exactly functionally equivalent and there is a very subtle difference between the two.

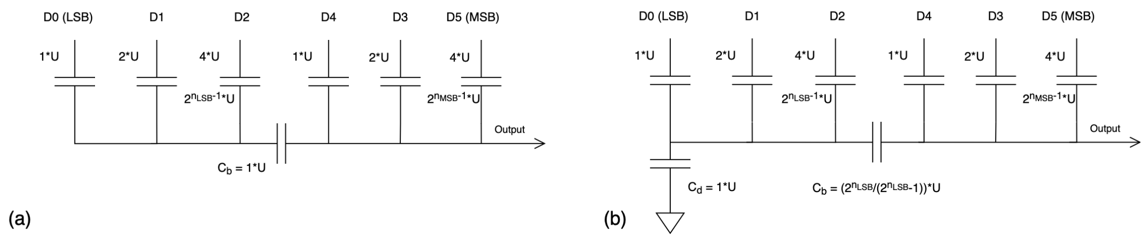


Figure 4: (a) Configuration without dummy capacitor. (b) configuration with dummy capacitor (C_d). Note that the MSB-LSB direction has been flipped compared to the CDAC in Figure 3. Here, $n_{LSB} + n_{MSB} = n = 6$ rather than 8 previously. Explanations of the formulae for (b) can be found in [7]

The subtle difference between the two is that without the dummy capacitor, the DAC relationship equation for Figure 4(a) is different than that of equation (2):

$$V_{out} = V_{in} \cdot \frac{k}{2^{n-1}} \tag{3}$$

Notice that we *can* output a maximally $V_{in} \cdot 63/63 = V_{in}$ for this 6 bit CDAC. With the dummy capacitor, the DAC relationship is the same as in Equation (2) and the maximum output is $V_{in} \cdot 63/64 < V_{in}$. Additionally, in either case, the bridge capacitor can be relocated asymmetrically such that n_{LSB} need not be equal to n_{MSB} [4]. Proving that these DAC topologies work is similar to that of in Figure 3. All that is required is redrawing the DACs. Generally speaking, these CDACs work by using the property that series capacitors reduce the overall capacitance; this can allow the LSB side to have smaller effective capacitance, preventing the need for larger MSB capacitors.

It is important to determine the tradeoffs and benefits of these two configurations. CDACs with bridge capacitors are smaller and have can less capacitance in general. Having lower capacitance is beneficial to reduce power consumption because each time a CDAC is sampling, the CDAC is being charged and discharged, and thus energy is lost given by ($\frac{1}{2}CV^2$) – a lower capacitance

[4] Multiple bridge capacitors can be included in configuration (b) to make a multisection CDAC or even C2C CDACs, and the mathematics for sizing capacitors are demonstrated in [7]. On an author’s note, I have experimented with adding more bridge capacitors to (a) and have been successful; however, I am not sure if there is a closed form solution to sizing the bridge capacitors and have not found any mention of this in literature.

proportionally decreases energy consumption. However, bridge capacitors may greatly harm the CDAC's linearity, critical to the ENOB in ADCs.

It is expected that parasitic capacitance will exist among all nodes during IC design and some topologies are more tolerant to these parasitic capacitance. To explore this, we label one side of a binary weighted CDAC the "top plate" (green) and the other side of the CDAC the "bottom plate" (blue). This naming convention will be explained later in 4.a.1. For simplicity and for purposes of illustration, we ignore all the parasitic capacitance from the driving digital logic nodes to the top plate. Since the bottom plate capacitors are actively being driven, having parasitic capacitance to ground is largely irrelevant (the voltages at those nodes are well defined by the drivers). However, the top plate is not being actively driven and the parasitic capacitance from the top plate to ground will affect the CDAC.

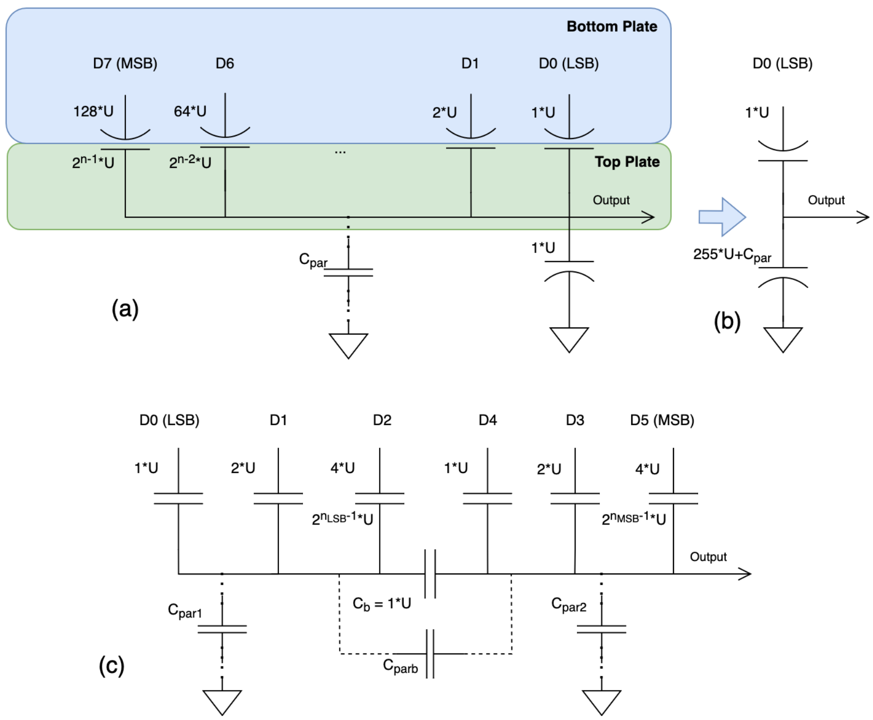


Figure 5: (a) Simple binary-weighted CDAC with parasitics. (b) Equivalent circuit to (a) when 8'b1 is input. (c) CDAC using bridge capacitor with parasitics.

A mathematical analysis will reveal that in Figure 5(a), all CDAC's output values will be scaled downwards linearly due to C_{par}. This is equivalent to multiplying the RHS of DAC equation (2) by some constant whose value is less than 1. Although this is undesirable for the *absolute accuracy* and will create gain error in the CDAC, this is not particularly detrimental to the *differential linearity* (all steps will be of equal size) and, thus, does not alter the ENOB of the CDAC. This is not the case in CDACs with bridge capacitors. Even if $C_{par1} = C_{par2}$, not all values will be scaled down linearly; some codes will be scaled down more than others. This gives rise to nonlinearity – there is no longer a linear input value/code to voltage correspondence. Some steps will be larger than other steps. This will adversely affect the ENOB of the CDAC, and thus affect the ENOB of SAR ADCs if not properly compensated for [5].

The C2S2 team has chosen to use a completely **binary-weighted CDAC** topology for its FA24SP25 ADC as:

1. We are not area constrained and thus do not need area savings.
2. We do not have a ADC with more than 8 bits.
3. There is less complexity associated with binary-weighted CDACs.
4. Parasitic management (and thus layout)is less complex with binary-weighed CDACs.
5. Although there is less power savings with binary-weighted CDACs (not very much due to it only being 8 bit), we value a higher ENOB more in this case.

3.b.2 Charge Redistribution CDAC

Another decision to make in the CDAC arena is between charge redistribution and charge sharing CDACs. Charge sharing CDACs are a relatively new low-power CDAC topology. In charge redistribution CDACs (like the CDACs shown in Figure 3 and 4), these CDACs require energy to change the output from one input code to another as capacitors are being charged and discharged. Charge sharing topologies take a different approach by precharging all capacitors at some ”initial” point in time (i.e. every time the ADC samples). The switching operations will merely connect and disconnect the already precharged capacitors such that each ”switching” operation to change the CDAC’s output voltage does not consume any more power [8].

Charge sharing topologies, although interesting and have useful benefits, are complicated and somewhat novel. Thus, C2S2 has opted to use the typical **charge redistribution** topology [5]. For the sake of brevity (CDACs are admittedly not the most interesting and this section is very long already), charge sharing topologies will not be discussed in detail and a great amount of information can be found in [8].

3.c ADC Timing Architecture

SAR ADCs need to compare multiple times to ”converge” on an output (the number of comparisons needed is related to the number of bits of the ADC). After each comparison, the output of the comparator needs to be fed back into a digital logic block to readjust the digital value to the CDAC and, thus, the CDAC’s output. This feedback cycle of comparison and CDAC switching (which we call a comparison-switching cycle here) can be self-timed or sequenced via an external clock.

[5] There are also some accuracy and complexity benefits with charge redistribution topologies (bottom plate sampling and no need for external sample and hold)

In a configuration sequenced by an external clock, the digital logic, which is a finite state machine (FSM), may be periodically updated on the rising edge of an external clock. The falling edge could also be used to clock the comparator, for example. In this configuration, the comparison-switching cycles is *synchronous* with an external clock. The external clock, is also incrementing a counter (which can be within the FSM) to track the number of comparisons to inform the ADC when to sample and when the comparison-switching operations have finished. For example, 8 clock cycles may be dedicated to comparison-switching and 1 clock cycle may be dedicated to sampling and outputting the previous converged value for an ADC that outputs every 9th cycle.

In a self-timed loop, rather than an external clock informing the FSM when to proceed forward, the comparator can ”tell” the digital logic that it is done comparing and achieved a steady state, and the FSM can step forward accordingly. Once the FSM has switched the CDAC and (after some pre-programmed delay), the FSM can ”tell” the comparator to perform a comparison. In the background, a counter is also keeping track of the number of comparisons, but this counter is not tied to any external clock. The only signal that selt-timed ADCs require is a ”go” signal that tells the ADC to sample, and the ADC will *asynchronously* step through the algorithm independently of any clock.

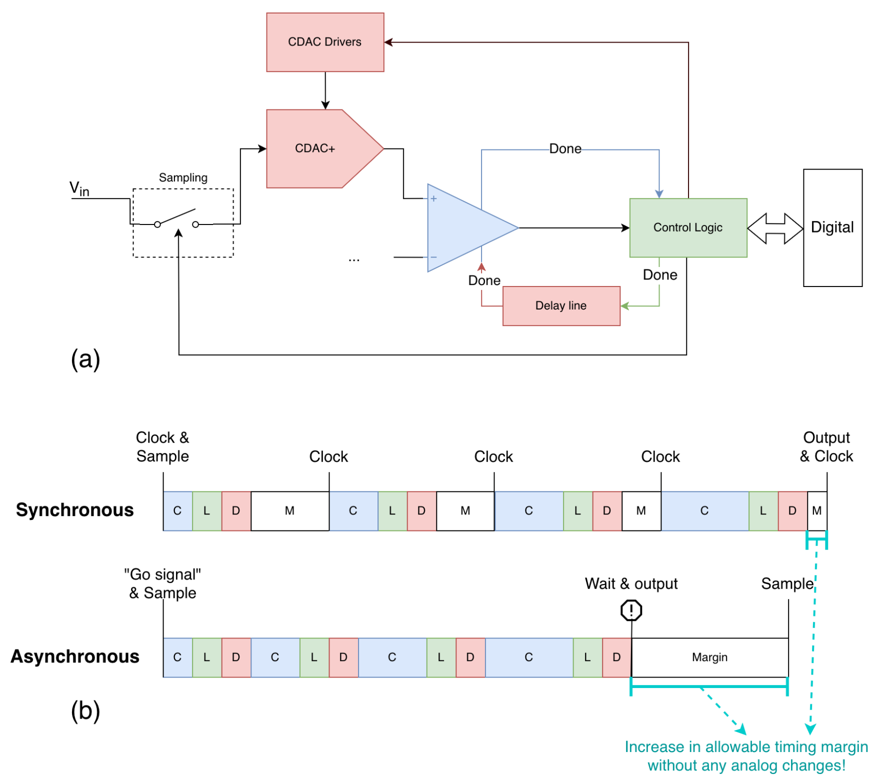


Figure 6: (a) Asynchronous ADC timing loop components. (b) ADC timing loop timing diagrams of each topology (graphic based on [9])

There are advantages and drawbacks with each. Timing wise, in the synchronous case, time will be wasted. The time the comparator takes to settle to a comparison is not constant (i.e. the propagation delay of the comparator is a function of the input voltages). Thus, a synchronous clock may barely meet timing for some comparison-switching operations and leave excess time on the table for other comparison-switching operations. In the asynchronous case, this problem is mitigated because the comparator will tell the FSM that it has finished comparing and all the timing margin can be summed together as in Figure 6. [9].

Design complexity wise, synchronous topologies are likely simpler to implement. Synthesis tools will require less setup to properly synthesize the FSM given timing requirements from a synchronous clock. Asynchronous topologies may require delay buffers and tuning to properly sequence the comparisons and switching while meeting timing. Creating a mixed-signal tapeout may be more challenging with asynchronous topologies as the "go" signal to valid output time delay could have a potentially unpredictable relationship to input voltage making the digital design more challenging.

In asynchronous topologies, delays are subject to process, voltage, and temperature effects (PVT). In the case that these delays are incorrect, it will be difficult or impossible to fix this after tapeout[6] – in the case of a synchronous ADC, the clock period may merely have to be increased if timing is not met.

Due to the potential for mixed-signal integration, low speed requirement and general emphasis on creating a minimum viable product, the C2S2 FA24SP25 SAR ADC uses a **synchronous** architecture. However, asynchronous architectures are definitely a priority item to implement in future tapeouts.

3.d Differential Architecture

ADCs (of any type) can be single ended or differential. Revisiting the balance/scale analogy, a differential architecture would be analogous to a balance scale with two unknown weights, and the difference between the weights has to be determined. Known reference weights can be added to either side of the scale to converge to a value (this hints that differential ADC architectures may have many different "algorithms" in converging, explored in 4.a.2).

In a single ended architecture, the ADC measures (quantizes and samples) the voltage of the input compared to some reference voltage, either V_{dd} or some external V_{REF}. In these ADCs, there is one V_{in}. In differential ADCs, there are two inputs to the ADC, V_{IN+} and V_{IN-}. The ADC will measure the *difference* between the two ($V_{IN+} - V_{IN-}$). There are many subcategories of differential ADCs including pseudo-differential, fully differential, and truly differential. Choosing between differential and single-ended architectures greatly influences ADC design, and in SAR ADCs, influences all components of the ADC: the sampling architecture, CDAC, logic, and comparator design.

[6] I have stumbled a paper or some ADC results, which I no longer find, where the ADC had to be tested at a low temperature to function properly because of this.

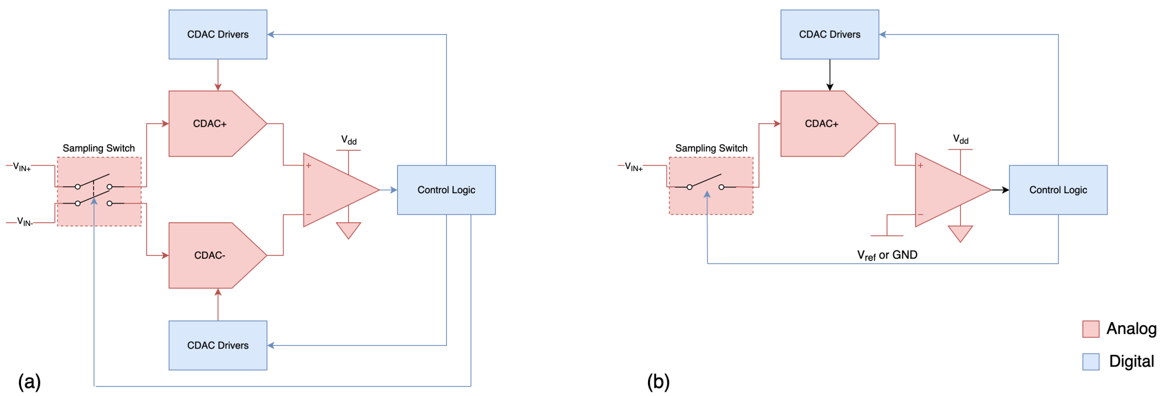


Figure 7: (a) Differential SAR ADC configuration. (b) Single Ended SAR ADC configuration.

Single ended ADCs are simpler in the digital logic design and require fewer components, but unfortunately, suffer from many limitations and greatly limits the possible design parameters.

Differential ADCs may ease comparator design. To understand this, a brief introduction into differential and common mode voltages and comparator functionality is introduced here. Comparators are inherently differential devices – they compare two voltages, even in a single ended configuration (apparent in Figure 7. The two voltages that enter a comparator can be decomposed into its common mode (V_{cm}), the average voltage between the two, and differential mode (V_{dm}), the difference between the two, components. V_+ is the voltage at the "+" terminal of the comparator and V_- is the voltage at the "-" terminal of the comparator.

$$V_{cm} = \frac{V_+ + V_-}{2} \tag{4}$$

$$V_{dm} = V_+ - V_- \tag{5}$$

Ideal comparators determine which voltage (V_+ or V_-) is larger, outputting a logic high (1) when $V_+ > V_-$ and a logic low (0) when $V_+ < V_-$. When $V_+ = V_-$, it does not output a valid logic level [7]. A different way of writing this inequality is when $V_{dm} > 0$ the comparator outputs a logic high and $V_{dm} < 0$ the comparator outputs a logic low.

In a comparator, the acceptable input common range is typically limited to smaller than from 0V to Vdd. They will not compare properly or compare very slowly (long delay before output settles to the correct value) if they are outside of this input range. The differential mode input range can usually be larger than this common mode input range. The limitations in comparator input voltage range (along with CDAC limitations) directly translates to ADC input voltage range. For the sake of discussing how the comparator limits the ADC's input voltage range, we ignore the CDAC and

[7] It depends on the comparator architecture as to what happens in reality here – clocked comparators will sit in a metastable state and eventually reach a logic value (like a flip flop) and static comparators will sit somewhere between the logic 1 and 0.

assume the output of the sampling switch directly passes into the comparator (we will explore how the CDAC nuances this).

It is not possible to keep the input common mode range for single ended ADCs constant since the comparator is being compared to 0 or some constant voltage. Substituting 0 or some other constant into V_- equation (4) will mean that V_{cm} will always depend on V_+. If the comparator's common mode range is $0.7V$ to V_{dd}, for example, the ADC will be limited to valid input ranges of $0.7V$ to V_{dd} too.

In differential mode ADCs, we can keep the common mode stable if the V_{IN+} moves "opposite" to V_{IN-} (i.e. any increase in V_{IN+} will also result in an equal decrease V_{IN-}). Since the differential mode voltage is changing but the common mode is not, we are not restricted to the common mode voltage limitation and only restricted by the differential mode limitation. The differential mode range can be $0 - V_{dd} = -V_{dd}$ to $V_{dd} + 0 = V_{dd}$ for example.

Furthermore, comparators function better with a stable common mode voltage. Although comparators should only be using the differential mode component to make the decision, in reality, the common mode component affects the output too. This is because real comparators are not perfect and have a finite CMRR (common more rejection ratio), which signifies how much the comparator can "ignore" the common mode and use only the differential mode component to determine its output. If we can ensure that the common mode voltage stays constant by using a differential mode ADCs, we can ease the CMRR design stringency and improve our full-scale input voltage range.

The comparators and sampling switches have other nonideal aspects such as kickback and charge injection (these will be discussed in 4.a.1 and in). Rather than needing these nonidealities to be very small (necessary in the case of single ended ADCs), these nonidealities merely need to be equal to each other on the differential inputs, and the differential nature of the ADC will subtract the nonidealities out.

Finally, but importantly, different sampling algorithms can be used with differential mode ADCs. Only 1 CDAC can be adjusted in single ended configurations, but differential mode implementations can adjust two CDACs independently and/or creatively within its sampling algorithm, saving power and improving accuracy.

The large benefits of differential ADCs greatly outweigh the difficulties from design complexity, which are mostly present in layout and the digital logic design, and possibly even reduce design complexity for the comparator. Thus, C2S2 has opted to implement a **differential mode ADC**.

4 Detailed Implementation and Operation Description

4.a Sampling and Switching Algorithm

The sampling architecture is one of the most important in SAR ADC designs. SAR ADCs all require some form of sample and hold circuit, whether that is incorporated in the CDAC or separate to the CDAC. As mentioned previously, C2S2 uses an architecture where the sampling mechanism is incorporated in the CDAC; the sampling mechanism can exploit that the CDAC has capacitance, which means that the analog voltage can be stored in the CDAC itself. There are two mechanisms in which a voltage can be sampled ”into” the CDAC: bottom plate sampling and top plate sampling.

4.a.1 Bottom Plate Sampling

In the capacitors used for this SAR ADC, there is an important distinction between the two plates (bottom and top plates) of the capacitor. Although on-chip capacitors are not polarized in the traditional sense (where you will damage them by applying a reverse polarity voltage), one side of the capacitor may not be functionally equal to the other plate of the capacitor. There are multiple types of on-chip capacitors including MOS capacitors, MOM capacitors, and MiM (Metal-insulator-Metal) capacitors. MiM capacitors will be of primary focus as these are used in the FA24SP25 CDAC.

In a semiconductor stackup (governing the layers of the fabrication process), one or more metal layers can be specially fabricated to be close to another metal layer, forming a parallel plate MiM capacitor, in processes that support MiM capacitors. In Figure 8 (b), the bottom plate parasitics to the substrate will be larger than the top plate’s parasitics as the bottom plate shields the top plate from interacting with lower metal layers and the substrate. Here, an asymmetry is present between the two plates as the bottom plate is more susceptible to parasitics and substrate interference than the top plate. Thus, the top plate is more appropriate for handling ”sensitive” signals and the bottom plate should be connected to a driven node (driven by a low impedance source), if possible. The bottom plates of capacitors are denoted with the curved end and the top plate with the flat end in the upcoming figures.

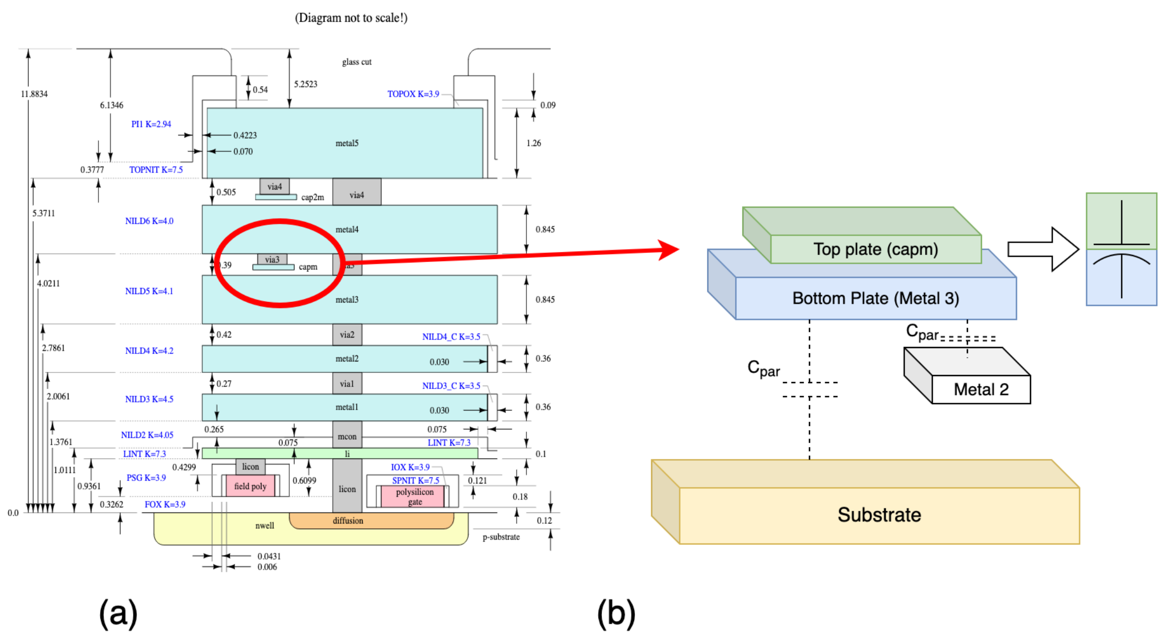


Figure 8: (a) SKY130 stackup incorporating 2 MiM cap layers. (b) Illustration of top plate and bottom plate in semiconductor stackup.

Another important nonideality to discuss in switches is the presence of charge injection. Every time an analog switch closes or opens, some charge will be added or removed to the sampling capacitor. This can be shown in Figure 9(b) below. Although the implementation in 9 (b) is naive, it demonstrates that switching the SW node results in charge leaving the sampling capacitor. Although the physical reason for charge injection is due to the channel of the MOSFET (and will be discussed briefly in the appendix), we model it here as nonlinear parasitic capacitances C_{par1} and C_{par2} [8]. When the switch is opened, charge leaves the sampling capacitor, reducing the voltage on the sampling capacitor and creating inaccuracy. We discuss this in more detail in 4.b.2.

What exacerbates this problem is that the charge injected onto the capacitor is *a function of the voltage on the capacitor (input voltage)*, creating nonlinearity [10, 11]. This is because the charge injected is not constant and a function of the voltage on the sampling capacitance (10. The input-dependent charge injection present in (b) can be reduced via bottom plate sampling.

[8] This model here is solely for simplicity. This model is actually not representative of what happens in charge injection and represents clock feedthrough [10]. Physically, the cause of charge injection is the channel charge dissipating

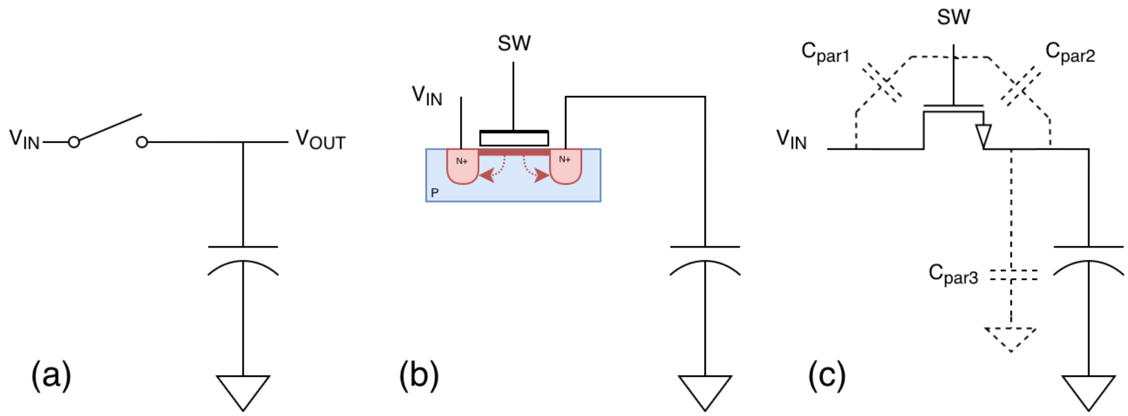


Figure 9: (a) Typical ”top plate” sample and hold circuit. (b) Physical representation of the mechanism of charge injection as part of a simple NMOS sample and hold circuit (c) A model, albeit unphysical, of charge injection as C_{par1} and C_{par2}. C_{par3} is a small parasitic capacitance from the top plate of the capacitor to the substrate.

A bottom plate sampling implementation (for a basic sample and hold circuit) is shown in Figure 10. The mechanism for bottom plate sampling is as follows:

1. When SW2 turns off before SW1 (T_1), charge leaves the top plate of the sampling capacitor due to charge injection from SW2 (I_2). This may affect V_{OUT}.

2. The top plate of the sampling capacitor effectively becomes open (apart from minimal parasitic capacitance from C_{par3}, which is small as this is the top plate) because SW2 has opened.

3. When SW1 opens and current (I_1) flows through C_{par2}, V_{OUT} is largely unaffected since little current flows through the sampling capacitor as the top plate of the capacitor is largely floating.

4. Although current/charge (I_1) will flow through C_{par4}, this will not affect the voltage across the sampling capacitor (V_{OUT}).

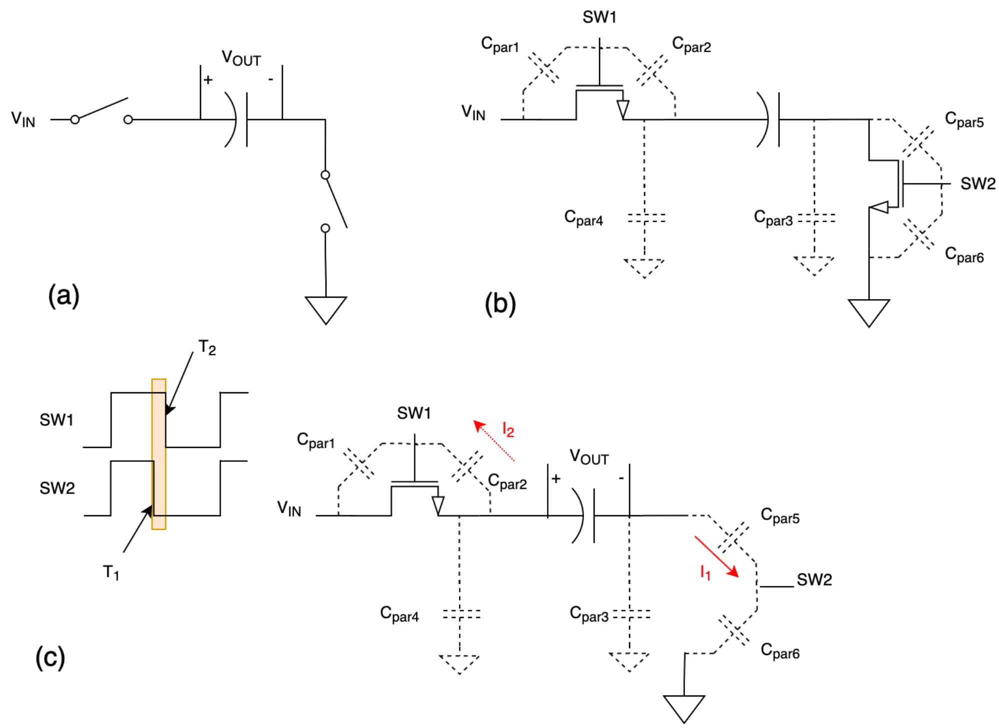


Figure 10: (a) Bottom plate sampling circuit (b) NMOS implementation with parasitics annotated (c) Switch timing and diagram showing switch states in the yellow region of the timing diagram.

Here, although there is some charge injection from SW2 through C_{par5}, this is much less problematic than in top plate sampling. The charge injection is *not a function of input voltage* and thus constant because the top plate of the sampling capacitor will be at GND, regardless of the input voltage before T_1 [11]. The differential topology is beneficial here because if the charge injection on both sides of the ADC is constant (the same) regardless of input voltage, there will be a constant voltage offset on each input, contributing only to a common mode offset. This will not affect the comparator output and thus not affect the ADC's reading.

The above diagrams represent a simple sample and hold architecture; some changes need to be implemented to combine bottom plate sampling and CDACs to eliminate the need for external sample and hold. To describe this, the figure below shows the transformation of top plate and bottom plate sampling into CDAC based sampling implementations.

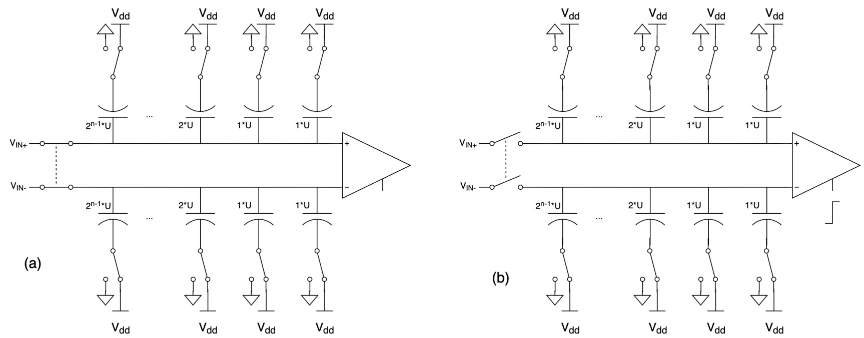


Figure 11: (a) Sampling operation in CDAC using top plate sampling approach (b) comparison operation in CDAC using top plate approach. [12]

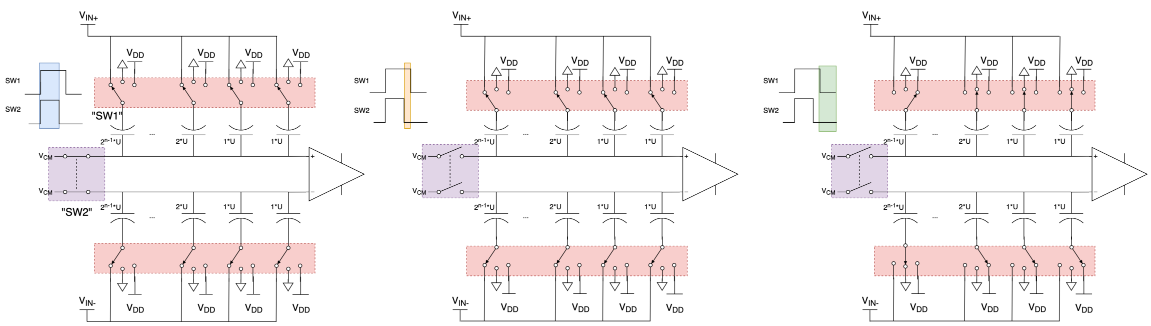


Figure 12: (a) Sampling operation in CDAC using bottom plate sampling approach (b) Bottom plate delay reducing charge injection (c) ”flipping” operation required for voltage readout/comparison. [12, 13]

Notice the similarities between the traditional sample and hold circuitry and those incorporating CDACs [9]. In the top plate approach, the sequence is as follows:

1. **Sample (a):** Connect the top plates of the positive and negative CDACs to V_{IN+} and V_{IN-}, respectively. Connect the bottom plates to V_{dd} or some other voltage reference.

2. **Compare (b):** The top plate of the positive and negative CDACs will be at V_{IN+} and V_{IN-}. The comparator can now compare. The CDAC inputs will also be disconnected from the input (input switches open).

3. **Switch:** The bottom plates of some capacitors will be switched to ground or some other voltage, according to the switching algorithm used. This will increase and decrease the voltages at the top plates of the CDACs.

4. The compare and switching operation will repeat until the algorithm has converged.

[9]This is a simple representation (using conventional switching) of top plate and bottom plate sampling CDACs; the specific switching algorithm used may change the CDAC further.

In the bottom plate approach, there are some additional steps to sampling. The sequence is as follows:

1. **Sample: (a)** Connect the bottom plates of the positive and negative CDACs to V_{IN+} and V_{IN-}, respectively. Connect the top plates to V_{CM} (not to be confused with V_{cm} previously). This will be a fixed voltage at usually $V_{dd}/2$.

2. **Bottom plate delay: (b)** The top plate of the positive and negative CDACs will both be at V_{CM}. This will make charge injection from SW1 equal on both CDACs when SW1 switches.

3. **Flip: (c)** A comparison cannot yet be done; both CDAC top plates are at the same voltage. To readout a voltage sampled by bottom plate sampling, the bottom plates are now switched to some constant known voltage (depends on algorithm). This will cause the voltage on the top plates of the CDACs to change as a function of the sampled V_{IN+}, V_{IN-}, and the known voltage.

4. **Compare:** A comparison can be done now as the top plate voltages are some function of V_{IN+} and V_{IN-}; this will inform a switching operation. Comparison and switching will repeat until the algorithm has converged.

Bottom plate sampling is used frequently in the design space for ADCs with greater than 10 bits of performance [12]. Although C2S2 does not require this, C2S2 opts to take this approach due to the minimal increase in complexity.

4.a.2 IMCS algorithm

Once sampling has been done, a capacitor switching algorithm will be used to adjust the CDAC such that the analog value can be determined. In a single ended SAR ADC, the options are relatively limited. However, differential ADC topologies have numerous options, which have all have distinct and complex tradeoffs. A designer has to carefully consider all of these tradeoffs. However, there is a substantial amount of literature detailing each switching algorithm, reducing the headache in choosing switching algorithms [8,13–15].

There are a few main factors in choosing which algorithms are suitable for a particular ADC[10]. These can include:

1. Power draw

2. The number of capacitors needed in the CDAC

3. Compatibility with bottom plate sampling

4. The need for extra voltages (e.g. V_{CM}) and the accuracy requirement for such voltages

[10] There are also algorithms incorporating redundancy in the ADC, reducing ADC errors [13].

5. The input voltage range the algorithm allows
6. Changes (if any) of the common mode voltage present at the comparator
7. The logic implementation complexity

The number of parameters increases the complexity of the design space when choosing a switching algorithm. Furthermore, the number of different switching algorithms in literature further increases complexity. Due to this and the abundance of literature, many of these algorithms will not be explored in detail here; a selected few algorithms that highlight the key differences between switching algorithms will be shown.

The conventional switching algorithm uses binary search. Although this is optimal in the case of a single-ended SAR ADC, this algorithm is a relatively power inefficient algorithm in differential topologies. Most basic depictions of SAR ADCs use the binary search algorithm as a simple way to describe their theory of operation; however, this algorithm is typically not the actual implementation in reality.

The conventional algorithm can be explained as the following diagram.

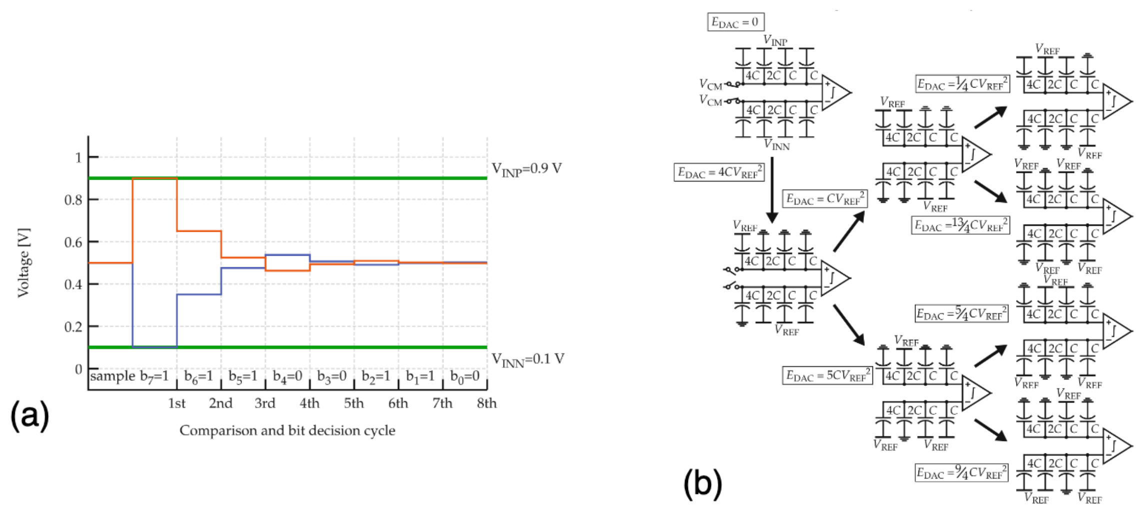


Figure 13: Figure screenshotted from [8]. (a) 8-bit conventionally switched CDAC top plate voltages. (b) 3-bit decision tree for conventional switching algorithm. In this diagram, V_{IN+} is denoted as V_{INP} and V_{IN-} is denoted as V_{INN}

The different stages are somewhat apparent. The very first stage in the tree is the sampling operation, and the second stage is the "flipping" operation. It is important to note that the flipping operation here connects the MSB capacitor to V_{REF}[11] and GND for the positive and negative CDACs,

[11] In this document, V_{REF} and V_{dd} will be used interchangeably as the FA24SP25 ADC sets $V_{REF} = V_{dd}$. In reality (and in the future), these nodes may be separate to allow for a much more accurate and clean V_{REF} as V_{REF} directly affects ADC accuracy/performance.

respectively. This is to prevent the top plate and bottom plate of the capacitors from immediately exceeding $V_{dd} = 2V_{CM}$ or sinking below 0V. After switching, a comparison is made; the result of this comparison governs which direction (arrow) the tree takes and the next switching operation that must occur. This occurs until all capacitors have been switched.

The voltage (ΔV_{top}) is defined as the change in voltage on the top plate of the CDAC from the previous voltage after switching. It will be a function of the charge into the top plate of the CDAC minus the charge moving out of the top plate of the CDAC. It is of particular interest to evaluate the flipping operation (the first arrow) of the example shown in Figure 13. ΔV_{flip} here is the change in the top plate voltage of the positive CDAC (CDACP) from V_{CM} after flipping for this specific case (i.e. V_{top} after switching equals $\Delta V_{flip} + V_{CM}$).

$$\Delta Q_{top} = Q_{in} - Q_{out} = 4C \cdot (V_{REF} - V_{IN+}) - (2C + C + C) \cdot (V_{IN+} - 0) \tag{6}$$

$$\Delta V_{flip} = \frac{\Delta Q_{top}}{C_{tot}} = \frac{4C \cdot (V_{REF} - V_{IN+} - V_{IN+})}{8C} = \frac{(V_{REF} - 2V_{IN+})}{2} \tag{7}$$

A similar analysis can be done for the negative CDAC (CDACN). Notice there are still voltage limitations. The following condition still must be met: $|\Delta V_{top}| \leq V_{REF}/2$ if $V_{CM} = V_{REF}/2$. Otherwise, the top plate voltage will go negative or exceed V_{REF}. This could damage the comparator following the CDAC. This directly translates to an input voltage limitation of $0 \leq V_{IN+} \leq V_{REF}$ (the negative CDAC would also have limitation $0 \leq V_{IN-} \leq V_{REF}$). This limitation discussed here is merely a simple example of a voltage limit imposed by a the first switching operation for one particular algorithm. Subsequent switching operations may impose further limitations on the input voltage. Other algorithms may also have different limitations; the input voltage limitation must be carefully considered for every switching cycle when choosing switching algorithms.

More generally, for other switching operations and other CDACs:

$$\boxed{\Delta V_{top} = \frac{\Delta Q_{top}}{C_{tot}} = \frac{C_1 \Delta V_1 + C_2 \Delta V_2 + ... + C_n \Delta V_n}{C_1 + C_2 + ... C_n} = \frac{\sum_{n=1}^{k} C_n \Delta V_n}{\sum_{n=1}^{k} C_n}} \tag{8}$$

The sign of the change in voltage automatically describes the sign of the charge, so there is no need to externally include negatives/subtraction. The change in Q is the weighted sum of the voltages moving in and out, where the weights are the values of the capacitors (For example $C_1 = 4C, C_2 = 2C$... and so on and so forth), and the voltages $\Delta V_1, \Delta V_2$... are the changes in voltage on the bottom plate of the capacitors corresponding to C_1, C_2...

Additionally, notice that ΔV_{top} is not a function of V_{CM}. The comparator detects differences of ΔV_{top} on CDACP and CDACN because the starting voltage is V_{CM} for both CDACP and CDACN and is treated as common mode voltage (ignored) by the comparator. This means that the absolute value of V_{CM} does not directly affect the accuracy of the comparisons and only affects the input voltage range and common mode voltage for the comparator (the input voltage range is, in fact, optimized at $V_{CM} = V_{REF}/2$). This voltage still, however, needs to be generated on chip or externally. Due to symmetry of the voltages in Figure 13 (a), it is apparent that the mean of the voltage on the top plates of CDACP and CDACN, which is the common mode voltage (V_{cm}) seen by the comparator, will always equal to V_{CM} (i.e. $(V_{topP} + V_{topN})/2 = V_{CM} = V_{cm}$). This means that the common mode voltage seen by the comparator (V_{cm}) can be fixed to a well-defined voltage (V_{CM}) that is acceptable to the comparator, reducing common mode input range limitations and CMRR limitations discussed in more detail in 3.d.

Finally, notice that on each switching operation, there are some instances where a capacitor is fully charged, then discharged again. This is very inefficient energy wise ($E = \frac{1}{2}CV^2$). A rigorous energy analysis (energy vs input) may be found elsewhere for this algorithm and thus is skipped [8,13–15].

It is important to note that some algorithms do not require an external V_{CM} for the algorithm to function. Some of these include the "energy saving" algorithm and the monotonic switching algorithm [6,13,14,16]. Although these will not be covered in detail here, there are some architectural advantages of not having a V_{CM} such as alleviating the need for a V_{CM} pin or an internal V_{CM} generator. Generating a V_{CM} may be challenging in some architectures which depend greatly on the accuracy of V_{CM}; in all architectures, it is necessary that the (on-chip or off-chip) V_{CM} is capable of driving switching capacitive loads[12].

Here, the FA25SP25 SAR ADC opts to use the IMCS (Inverted Merged Capacitor Sampling algorithm) [13]. This algorithm has a few key benefits over conventional switching including fewer capacitors necessary for the CDAC and greatly reduced power consumption when converging. Some positive features of the conventional algorithm are retained such as constant common mode voltage present seen at the comparator and bottom plate sampling. The IMCS algorithm does, similarly to the conventional algorithm, require a external V_{CM} during the switching and sampling operations; however, the accuracy of $V_{CM} = V_{dd}/2$ is not critical to the performance of the ADC and has good tolerance to variation similar to the conventional algorithm.

The switching sequence of the IMCS algorithm is clearly explained in the following diagram:

[12] This means that the buffer driving the V_{CM} may have to be a unity-gain open loop buffer or unity-gain stable in feedback.

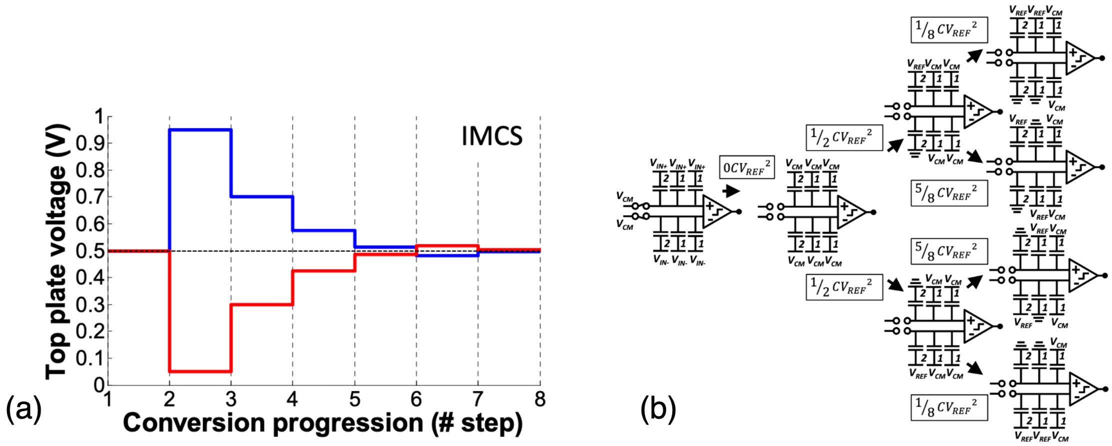


Figure 14: Figure Screenshotted from [13]. (a) 6-bit Inverted Merged Capacitive Sampling CDAC top plate voltages. (b) 3-bit decision tree for IMCS algorithm.

One **critical** aspect of this algorithm is an n-bit IMCS ADC requires only an $n-1$ bit CDAC. In this particular example, notice that the IMCS 3-bit ADC uses a 2-bit CDAC rather than a 3-bit CDAC present in the conventional switching algoirthm. In other words, the IMCS reduces the number of capacitors required, compared to conventional switching, to create an n-bit ADC by an entire 2^{n-1} for a binary weighed CDAC. This is not only a property of the IMCS algorithm, but a handful of other algorithms too. This greatly saves on CDAC layout design complexity and energy consumption.

Similar to the conventional switching algorithm, the IMCS algorithm suffers from input voltage limitations too. Previously, one of the voltage limitations of V_{IN-} and V_{IN+} were discussed. Both of which have to be within 0 and V_{REF}. However, there is another more nuanced limitation present for many ADC switching algorithms, which will be discussed in 4.a.3. Before this discussion, the actual implementation for the FA24SP25 tapeout will be shown.

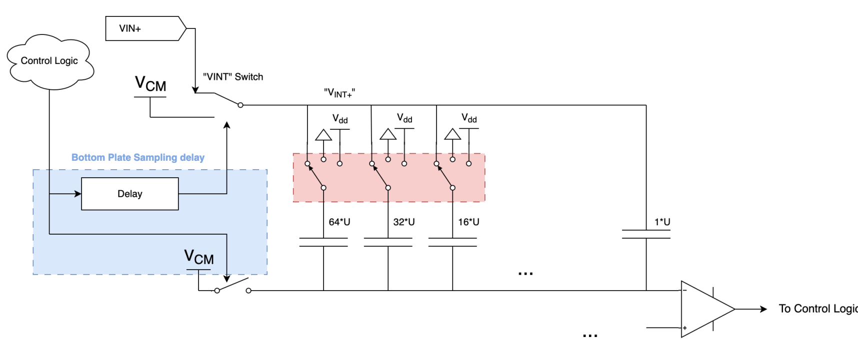


Figure 15: Overview of Implementation of IMCS in FA24SP25 ADC

Here, the implementation of the IMCS algorithm in the FA24SP25 ADC is shown. Only the positive CDAC (CDACP) is shown for brevity. The analog switches are controlled by the control logic as well (not included on the diagram for organization). The implementation details of the analog switch and digital logic will be discussed in more detail in the relevant sections. However, the bottom plate sampling delay can be seen (which is implemented with a delay standard cell).

In the IMCS algorithm (focusing on the positive CDAC) in Fig. 14, the bottom plate of every capacitor must be able to connect to 4 different voltages (V_{dd}, GND, V_{IN+} and V_{CM}). However, V_{IN+} is only needed during the sampling phase and is no longer used in the switching/comparison cycles. Each analog switch connected to the CDAC (the CDAC drivers) has 3 positions that it can connect to: V_{dd}, GND, and a node (denoted as V_{INT+}) connected to another switch (VINT switch) that toggles between V_{IN+} and V_{CM}.

1. In the sampling stage, the VINT switch is connected to V_{IN+} and the CDAC drivers are connected to V_{INT+}.

2. During the flipping stage, the CDAC drivers remain connected to V_{INT+} and the top plate V_{CM} switch opens. Due to bottom plate delay, VINT switch is connected to V_{CM} after the top plate V_{CM} switch opens.

3. Afterwards, for the switching-comparison stages, the VINT swich remains connected to V_{CM} and the CDAC drivers toggle according to the IMCS algorithm.

4.a.3 Differential Implementation and Fully-Truly switch

With some combinations of input voltages using the IMCS algorithm (and most other algorithms), the voltage at the top plates of the CDACs (and thus the inputs of the comparators), may exceed the power supply rail ($0 < V < 1.8$ in the process used by the FA24SP25 tapeout). The act of such voltages exceeding the power supply rail is known somewhat well as the ”over-range” problem [13]. There are many ways of mitigating this, one of those is bounding the voltage even more than previously shown. The acceptable input range for the IMCS algorithm can be bounded to the following:

$$\boxed{\frac{V_{IN+} + V_{IN-}}{2} = V_{cm} \approx V_{REF}/2 = V_{CM}} \tag{9}$$

This equation essentially means that the common mode component of the inputs should not deviate from $V_{REF}/2$, which is typically the externally supplied V_{CM}. This mode is also called the *fully-differential mode* [17, 18]. Otherwise, this could cause damage to the ADC. The derivation of this equation is done in detail in the Appendix (6.a.1). However, the main takeaway is that the designer has two choices if the entire input voltage range is to be retained. The designer can impose the aforementioned limitation or modify the CDAC.

Voltages not satisfying this criterion would result in voltages outside the acceptable power supply range and may lead to damage of the comparator or other circuitry. However, this restriction may not always be viable. Especially upon power-on, the voltages of V_{IN+} and V_{IN-} may both be at 0V. Thus, there is motive to create a *true*-differential mode [17, 18], which would allow arbitrary inputs between 0 and V_{REF}.

To avoid this, extra "grounding" capacitors can be added to the CDAC for "swing reduction" [13, 19]. Evidently, "swing reduction" reduces the amount of voltage that the CDAC can swing around V_{CM}. This inherently causes difference the voltage difference between CDACP and CDACN to be smaller, forcing the comparator to compare smaller differences in voltage compared to operation without these capacitors. C2S2 has opted to add a switch to the capacitor, allowing both modes to be use. The default mode is with the capacitors connected upon power-up until a certain mode select pin is pulled high.

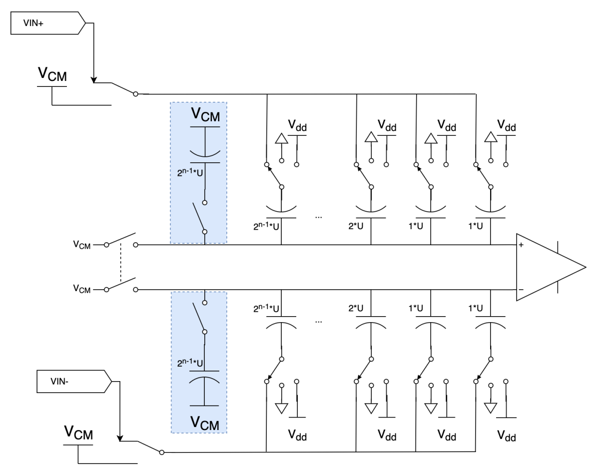


Figure 16: The swing reduction capacitors and switches to connect/disconnect these capacitors are highlighed in blue. This figure depicts the implementation of swing reduction capacitors in conjunction with the IMCS algorithm used in the FA24SP25 tapeout.

4.b Switches and Drivers

4.b.1 Driver Module

The CDAC bottom plates need to have voltage applied to operate it. The inputs of the CDAC present a relatively large capacitive load (hundreds of fF to several pF) to the stage driving it. For this reason, the control logic cannot directly drive the CDAC as the preceding digital gates (chosen by a synthesis algorithm) may not be designed to drive a large capacitive load. Furthermore, the sampling rate of the ADC is, in part, determined by the settling time of the CDAC – this is dependent on the time it takes to charge the capacitors in the CDAC. A specialized driving circuitry needs to be designed to carefully control the CDAC settling time without needlessly spending power and impacting accuracy.

In Figure 15, the switches highlighted in red are merely a functional representation the CDAC driver and are not actually the true design of the CDAC driver. The true CDAC driver use in the FA24SP25 tapeout can be seen in the figure below (Figure 17).

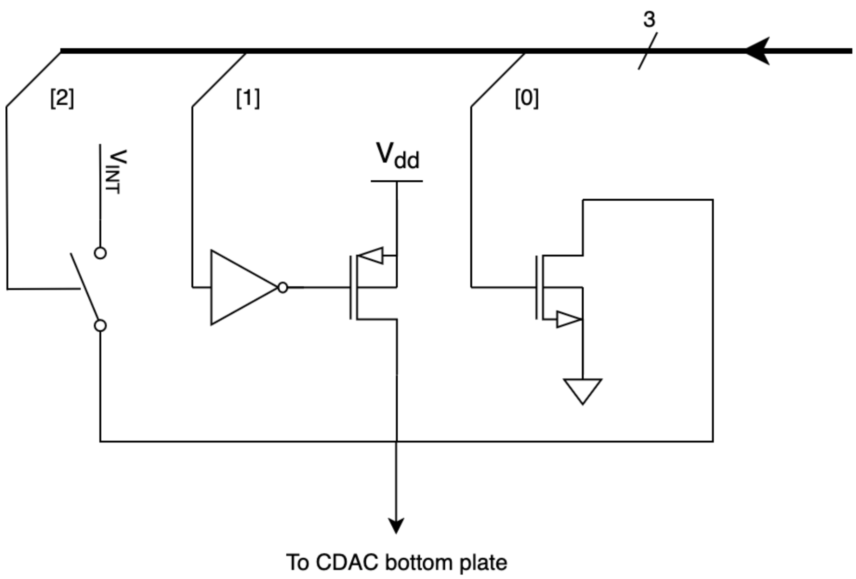


Figure 17: CDAC Driver implementation for FA24SP25 ADC. The V_{INT} connection connects to V_{INT+} or V_{INT+}, labeled in Fig. 15. The 3 bit bus comes from a preceding decoder; only one of the inputs should be high at one time.

Rather than using 3 seperate analog switches in implementing the CDAC driver, NMOS and PMOS transtors are used to pull to GND or V_{dd} or V_{ref}, respectively. NMOS transistors are not very effective in pulling to V_{dd} and PMOS transistors are not very effective in pulling to GND. Using only one transistor rather than an analog switch reduces charge injection and complexity. This is not the case for V_{INT} as this node may be at some voltage between V_{DD} and GND, and thus requires both transistors to pull up and down. This will be explored in detail soon.

The same driver cannot be used for every input of the CDAC. Depending on the capacitor that is being driven in the CDAC, the NMOS and PMOS will need to be sized differently. In particular, the width of the NMOS and PMOS will need to be different. Notice that in all the CDACs (Figure 15, 3, 4 etc...), each input has an different value capacitor connected to it. Generally (in binary weighted CDACs), the MSB input "sees" effectively the most capacitance and the LSB input sees the least capacitance. Intuitively, the driver that drives the MSB input has to be designed differently than that driving the LSB input.

As the transistor width is increased, proportionally more current is allowed to flow through that transistor (this is an oversimplification, but adequate for this discussion). If a designer wanted to charge or discharge a large capacitor in the same time it would take to charge or discharge a small capacitor, a wider transistor is needed for the large capacitor compared to the small capacitor. Thus, to drive the MSB, a wider NMOS and PMOS are chosen than that for the LSB. These sizing specifics will be discussed next in the Analog Switch section.

4.b.2 Analog Switch

In a high-level perspective, an analog switch essentially acts as a mechanical switch or relay, but is electrically controlled. Specifically, an arbitrary (analog) current is allowed to pass bidirectionally when the switch is closed (the switch acts like a wire) and no current passes when the switch is open (the switch acts like an open-circuit). In reality, analog switches have numerous parameters that need to be considered including on resistance, on resistance nonlinearity, charge injection, and leakage.

A diagram of a basic, canonical, analog switch (also called transmission gate) is shown below:

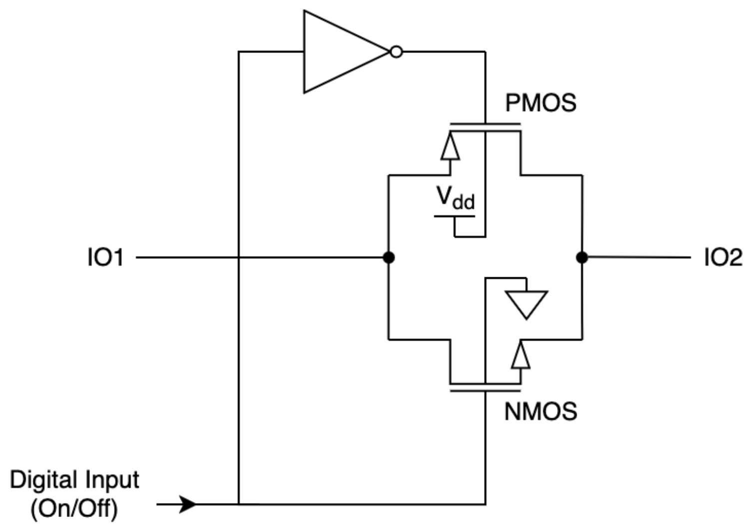


Figure 18: Basic Analog Switch. Although the NMOS and PMOS have source and drain drawn, these transistors are symmetric and source/drain do not have a clear distinction.

A simple example can demonstrate the working of an analog switch (while also ignoring the details of MOSFET functionality). Assume in this case that IO2 is connected to a discharged capacitor (whose other terminal is connected to 0V) and IO1 is connected to 1V. Ignoring the PMOS, an NMOS transistor conducts when its gate voltage is larger than its source voltage by some threshold $V_{TN} = 500mV$ and thus when the digital input is high (1.8V), current flows to IO2. This will occur until the voltage at IO1 equals that of IO2.

However, assume that now the voltage of IO1 is at 1.8V and the digital input is high (1.8V). Again, disregarding the PMOS, the current will flow to the source until IO2 reaches $1.8 - V_{TN} = 1.3V$. If IO2 charges to be any higher than that, the NMOS transistor will turn off. This is an issue because $IO1 \neq IO2$ when the switch is fully on/closed, which does not satisfy the operation of a switch.

Thus, the PMOS transistor is added to assist the NMOS. A PMOS transistor turns on when its gate voltage is lower than its source voltage by some threshold V_{TP}. If IO1 is at 1.8V and the digital input is high, the input to the PMOS will be low at 0V (due to the inverter). This means that the PMOS will be fully on and can charge the capacitor at IO2 to the required 1.8V.

Flipping the analog switch should yield the same results. To explore this, assume now IO2 is a charged capacitor at 1.0V and IO1 is at ground. At first glance, interpreting the transistors' states are difficult. However, by definition of "source" and "drain", the carriers (electrons for NFETs and holes for PFETs) of the MOSFETs always enter the FET via the "source" and leave via the "drain", determined by the gate [20]. Thus, the MOSFET's source and drain can depend on the voltages around it – this means that the drain is typically defined to be positive relative to the source in symmetric N-channel FET [21] (and vice versa for symmetric P-channel FET) [13]. Thus, to analyze this case, the drain is now assigned to the IO2 side for the NMOS and the IO1 side for the PMOS. Then, current can flow from IO2 to IO1 as both transistors will turn on until the capacitor is completely discharged.

As discussed previously, the switch should act like a wire when it is closed and open when it is open. In real devices, the switch has an effective resistance across it, which is dependent on a variety of factors relating to the transistor, and even the input voltages. This value of resistance is typically denoted R_{dson} Although the derivation of the resistance across the MOSFET requires more in-depth knowledge of transistors (which would be too lengthy to discuss here), at first order, **the resistance R_{dson} is proportional to the length of the MOSFET and inversely proportional to the width of the MOSFET.** Thus, wider transistors (longer drains and sources) would result in less R_{dson} and longer transistors (more distance between drain and source) would result in more R_{dson}.

Lumping the analog switch effective resistance as R_{on} is a bit of a misnomer. This is because the value of R_{on} is *variable for different input voltages*. To explore why this is the case in detail requires prerequisite understanding of MOSFET fundamentals (which is explored in 6.b.1). However, the previously shown analog switch resistance varies according to the input voltage as the following:

[13] This is only the case because on most typical CMOS processes, the transistors are symmetric (look for a cross section of a MOSFET!). This is not the case for power FETs and LDMOS devices (the sources and drains are not interchangeable).

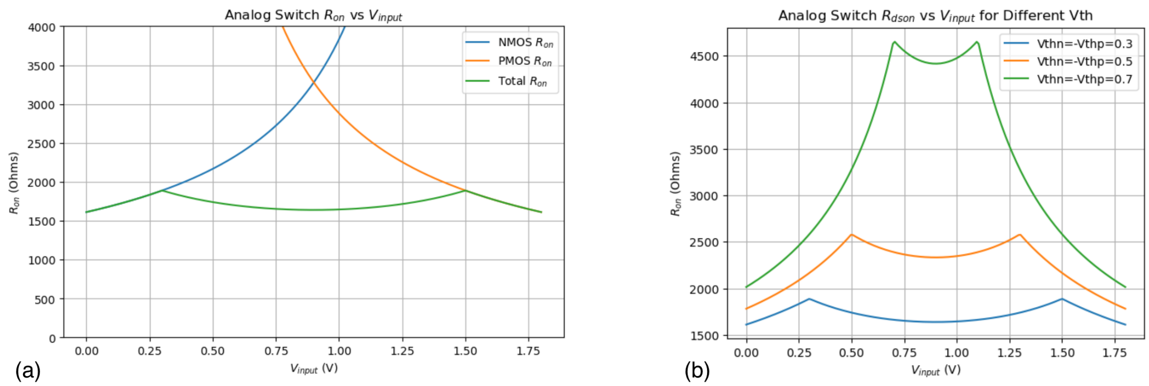


Figure 19: (a) Analog switch R_{on} vs V_{input} with NMOS and PMOS contributions labeled. (b) Analog Switch R_{on} vs V_{input} against varying threshold voltages.

Each analog switch in a SAR ADC should be sized according to the load capacitance/CDAC weight that it is driving. One may question the reason for sizing each transistor accordingly instead of choosing a large transistor that meets the charge/discharge time requirement for the largest capacitor as a ”one size fits all” solution. Charge injection (Q_{inj}) limits the size of switches. The wider a transistor is, the more capacitance the transistor gate has to the drain and source (i.e. the more C_{gs} and C_{gd} there is). Charge injection (especially charge injection from when the switch opens), would add excess charge onto the bottom plate of the CDAC after sampling has happened, contributing to a voltage error[14].

It is therefore useful to minimize the size of switch (and minimize Q_{inj}) from the analog switch in accordance to the load that it is driving. To approach this sizing, we can model the CDAC and CDAC driver with a simple RC model. Each of the inputs of the CDAC is treated as a capacitor, and each of the switches/drivers has a characteristic R_{on}. We determine the sizing of the switches/driver transistors by finding the maximum R_{on} that still continues to meet a settling time constraint, defined by the designer.

[14]This is also called pedestal error in literature [22].

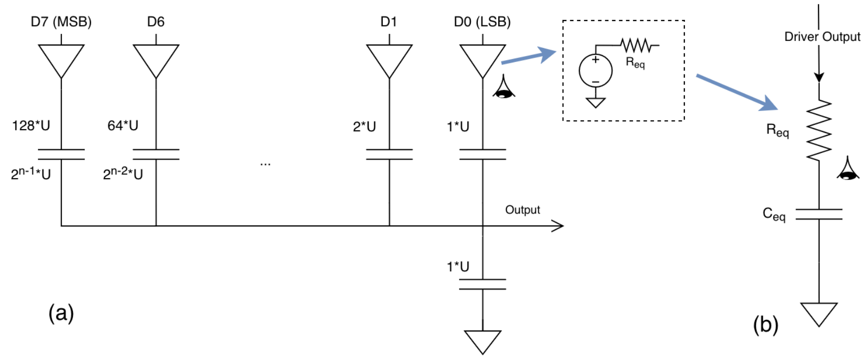


Figure 20: (a) Binary Weighted CDAC with drivers (b) equivalent capacitance seen by a driver looking towards the capacitor it is driving. For an 8-bit CDAC driver driving the LSB, $C_{eq} = \frac{1 \cdot 255}{1+255} \cdot \mathrm{U}$

The settling time constraint can found with this model and specifications, including the reference voltage, the resolution, and the desired speed. As RC circuits have exponential behavior, we know that if a step is prescribed to an RC circuit, the voltage at the capacitor will reach 63.2% after 1τ, 86% after 2τ, 95% after 3τ etc..., where $\tau = RC$. More generally, $\%Reached = (1 - e^{-t/\tau}) * 100\%$. In an ADC, it is necessary that the CDAC voltages settle to a voltage less than 1 LSB away from the reference voltage in the desired amount of time as defined by the speed. This must be the case to accurately resolve down to 1 LSB. In an 8 bit ADC, 1 LSB is approximately $1/256 = 0.4\%$ of the maximum/reference voltage. Therefore, it is necessary that the percentage reached is $100\% - 0.4\% = 99.6\%$, which corresponds to around 6τ [15].

It is apparent that in order to decrease the settling time of the DAC (which is required in high speed ADCs), either resolution must be sacrificed or τ must be made small. As C is defined by the unit capacitor size and the CDAC array topology, it is preferable, in high speed ADCs, to use small unit capacitor values and split-array CDACs. Sometimes, the C value cannot be changed so the R value must also be small enough to meet the timing requirement. This sets the maximum R_{on} of the driver and thus analog switches as part of the driver [16].

As mentioned previously, the R_{on} of the switch is set by the length and width of the transistors (wider transistors yield lower R_{on}). Unfortunately, wider transistors also have more capacitance and channel charge, and thus more charge injection; this leads to another tradeoff. The lower bound on $R_{on}$is governed by the maximum tolerable charge injection. When charge is injected on the sampling capacitor or CDAC input, it will cause $\Delta V = Q_{inj}/C$ change in voltage. If the Q_{inj} is larger than an LSB, it may cause measurable error. The tradeoff between R_{on} and Q_{inj} therefore imposes a limit on speed and resolution.

[15]In the FA24SP25 ADC, double the time is actually required because only half a clock period is allowed for settling, which is admittedly inefficient.

[16]This also indicates importance of R_{on} nonlinearity. A highly nonlinear switch will lead to varying charging times for varying input voltages (the CDAC will charge slower at some input voltages versus others), manifesting as nonlinear ADC operation

A method to somewhat alleviate this tradeoff is using charge injection cancellation devices. The charge injected onto IO1 by the transmission gate is assumed to be equal to the charge injected onto IO2 [17]. Given such assumption, the charge injected here can be roughly expressed as [10]:

$$Q_{inj} = \frac{WLC_{ox}(V_{DD} - V_{in} - V_{TH})}{2} \tag{10}$$

As the charge injection cancellation devices have both drain and source connected to the transmission gate, they impart $2Q_{inj}$. Thus, in order to effectively cancel the charge injection, they are sized with the same L but with half the W as the transistors in the transmission gate. Specifically, the compensation PFETs are half the W of the PFET in the transmission gate, and the compensation NFET is half the W of the the NFET in the transmission gate [18].

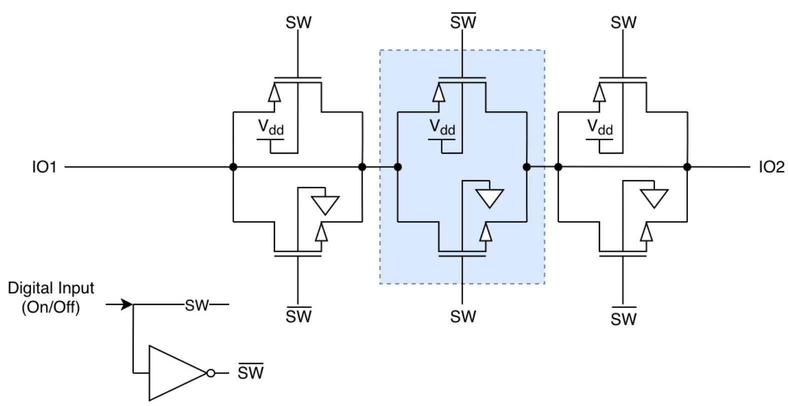


Figure 21: Diagram of analog switch with charge injection cancellation devices. The transmission gate (which performs the switching action) has transistors highlighted in blue. The non-highlighted transistors are charge injection cancellation devices.

Conceptually, charge leaves through the transmission gate NFET due to the falling edge action of SW. This is canceled by the due to the rising edge action of $\overline{SW}$ imparting charge through the cancellation NFETs. This happens similarly with the PFETs.

4.b.3 Break Before Make

For sensitive analog signals, two analog switches should not be on at the same time, even for a short period of time. A real-world application/instance where this can occur is the "VINT" switch in Figure 15. This switch is an SPDT (single-pole dual-throw) switch, meaning it connects one terminal between two other analog switches. To implement this SPDT switch generally requires

[17] This is not necessarily the case and may depend on a variety of factors such as impedance at IO1 and IO2. We also ignore body effect, which creates charge injection nonlinearity [10].

[18] A curious reader may question why the charge injection of the NFET does not cancel the charge injection of the PFET in the transmission gate (and vice versa). They would cancel if they were sized the same, but they are typically sized differently for lower R_{on} nonlinearity (due to electron and hole mobility differences)

two SPST (single-pole single-throw) switches, whose control terminals are inverted. The SPST switch is typically just an analog switch (or some variant thereof) like previously mentioned. A diagram of an SPDT implementation using SPST analog switches is shown below.

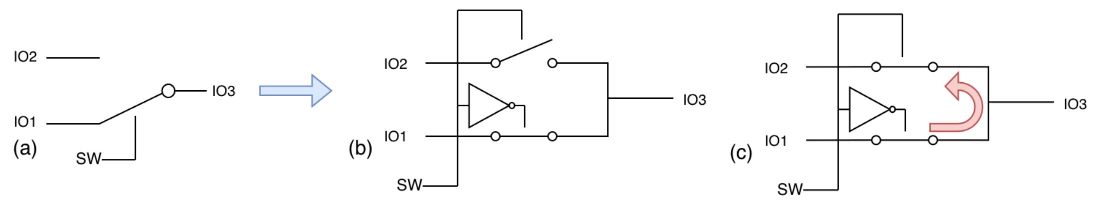


Figure 22: (a) SPDT switch (b) SPDT switch implemented with SPST switches (c) Shoothrough

Unfortunately, this SPDT switch currently does not possess a ”break before make” characteristic. This is because the inverter has some finite delay which causes both switches to be on at some point in time and causes ”shoot-through” between IO1 and IO2. This can cause undesirable switching transients, especially if IO1 and IO2 are sensitive analog signals. To remove this undesirable byproduct, a specialized timing module has to sequence the switching of the first switch and second switch. Specifically, some ”dead-time” has to be added such that there is a point where no switch is on (both are disconnected).

The FA24SP25 SAR ADC uses two types of break before make modules: a 2 terminal and a 3 terminal variant. The two terminal break before make is also commonly found in literature as a non-overlapping clock generator [23]. Due to the abundance of literature outlining its functionality, we skip such discussion here. It is used in places where SPDT swtiches exist in the ADC, such as the aforementioned VINT switch.:

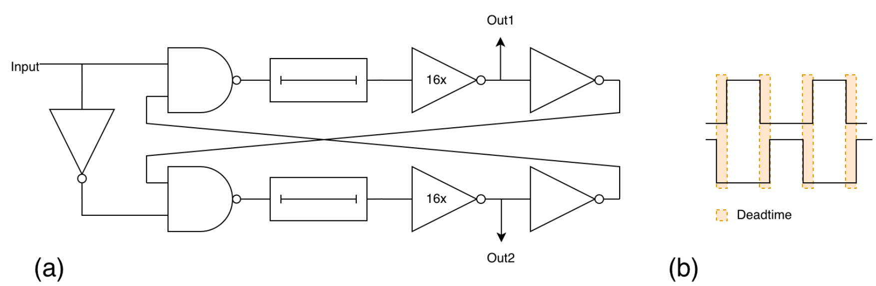


Figure 23: (a) Two terminal break before make (non-overlapping clock generator) (b) Non-overlapping output with dead-time labeled.

The 3 terminal break before make module is used to control the driver (shown in 17). The driver needs generate 3 non-overlapping signals. The module takes 3 input (the output from a digital decoder which is not guaranteed to be non-overlapping) and makes it such that only 1 output will remain on at all times. The module to do this is shown below.

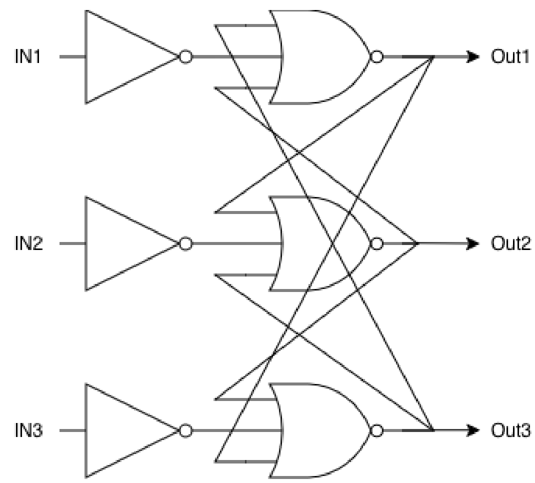


Figure 24: 3-input 3-output break before make module

4.c Comparator

Generally, a comparator compares the analog voltage at one node to the analog voltage on a different node. If the voltage for one node is larger than the other, it outputs a certain digital state (and vice versa for the other case). If the voltages are exactly equal (which is never the case in real analog circuits due to noise), then the output is indeterminate.

It is important to realize that comparators are essentially differential amplifiers with very large gain that clip the output to between VDD and GND (also known as a limiting amplifier). This is a useful idea to keep in mind while designing comparators. Despice the simplicity of this, comparator design is definitely non-trival and this upcoming section will only mostly provide a conceptual introduction to comparator operation.

A few key specifications of comparators that will be briefly touched upon in this section are the following:

1. **Speed:** the amount of time the comparator takes to make a comparison
2. **Power consumption:** the amount of power drawn by the comparator, on average
3. **Common mode input range:** the range of acceptable common mode voltages where the comparator can still operate
4. **Common mode rejection ratio:** the ability for the comparator to ignore fluctuations in common mode voltage
5. **Kickback:** charge that is re-imparted on the input during comparisons, affecting comparator accuracy

6. **Offset:** circuit performance/accuracy variations due to manufacturing process imperfections

7. **Noise:** random fluctuations (due to physics) that alter the circuit performance

One classification of comparators that can be made is clocked (dynamic) versus un-clocked (static) comparators. A static comparator is continuous time. Any moment that the positive input of the comparator is larger than the negative input, the comparator outputs a digital high; otherwise, it outputs a digital low. A dynamic comparator is discrete time – it will only compare at certain points in time, such as the rising edge of a clock signal. When the comparator is not being asked to compare (i.e. not on an edge), it holds the last comparison until it is asked to compare again. This is illustrated in Figure 25.

In cases where a clock signal is available, dynamic comparators have a few large advantages over static comparators. Most significantly, dynamic comparators can have lower power draw as they can be designed to only draw power when requested to compare. Additionally, they are typically faster as they can incorporate positive feedback/regenerative gain [24]. As SAR ADCs (even asynchronous ones) often make a clock/strobe signal available, it is often to the designers' advantage to use a dynamic comparator. A dynamic comparator is used in the FA24SP25 ADC.

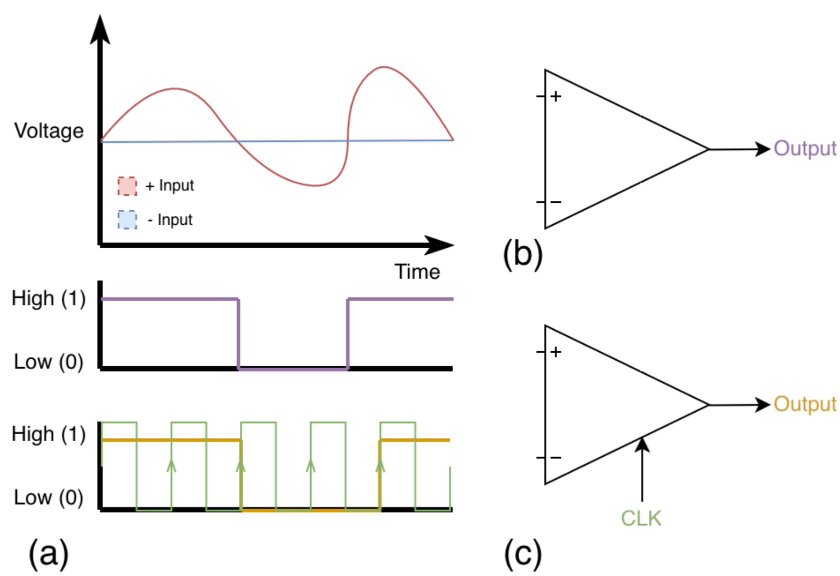


Figure 25: (a) Waveforms of inputs outputs of (b) un-clocked/static and (c) clocked/dynamic comparators

4.c.1 Regeneration

Within the clocked comparator space, there are a variety of amplifier topologies including StrongArm, double-tail, triple-tail, class AB, and more. These comparator architectures typically have one significant item in common, a stage that provides positive feedback/regenerative gain. Four representations of positive feedback that highlight the benefits of positive feedback are shown below:

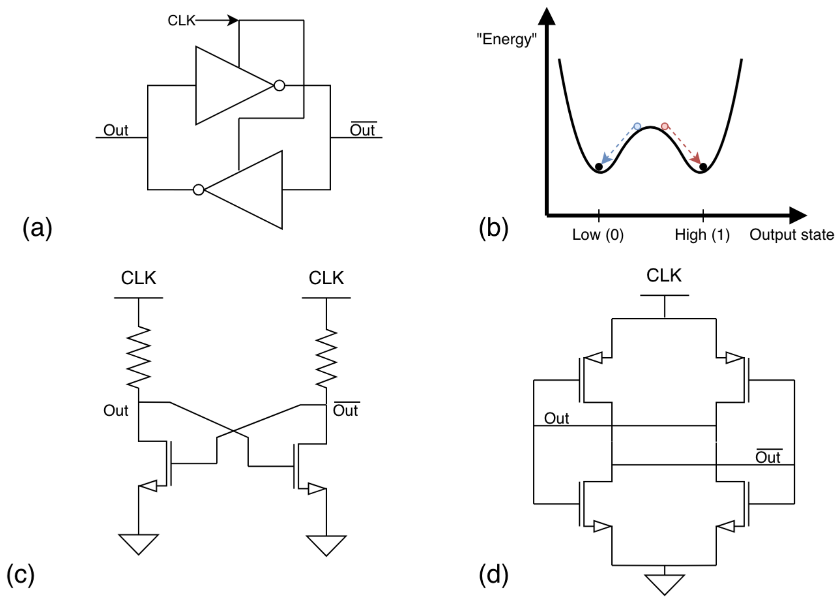


Figure 26: (a) Common building block for positive feedback seen in many comparators. (b) Physics-minded representation of positive feedback. Resistively loaded (c) and CMOS (d) implementations of (a) are shown.

A digitally minded person may immediately recognize (a) as being similar to an SRAM cell, which is indeed the case, except that there is a clock. The clock modification ensures that the two inverters are powered off when the clock is low.

A physics minded representation of circuit (a) in figure 26 is shown in (b)[19]. When clock is low, the circuit can be thought of as a potential with a single well (not illustrated); it is apparent that both outputs are low when the circuit is powered off/clock is low. When the clock turns on, the circuit has now two energy potentials (illustrated in (b)); the circuit has two stable states and 1 metastable state. The initial states that $\text{Out}_{t=0}$ and $\overline{\text{Out}}_{t=0}$ are at before the clock goes high determines the behavior of the circuit. If the initial conditions are that such the circuit slightly favors the Out=Low state ($\text{Out}_{t=0} < \overline{\text{Out}}_{t=0}$), then the circuit will "regenerate" until Out reaches the stable low state (Gnd) and $\overline{\text{Out}}$ reaches Vdd. This is illustrated by the blue ball and arrow. The opposite initial condition will result in Out regenerating to the stable high state.

The key takeaway is that in circuits with positive feedback, such as the one shown above, a small difference in inputs/initial conditions will cause a large amplification/large change in outputs after enough time has passed. It is guaranteed that after enough time, one of the outputs will be railed to VDD and the other will be railed to GND – this is exactly what is wanted in a comparator.

[19] This type of representation is also useful in discussing adiabatic logic, outside the scope of this document

Conceptually, ignoring the mathematical derivation, the regeneration stage speed (focusing mainly on 26 (d)) increases with gain of the transistors and decreases with capacitance on the output nodes [25].

4.c.2 Preamplification

In the previous regenerative circuits, there is no dedicated input – the input is also the output. This is undesirable as the regenerative stage will alter/pull the input [1]. Thus, a buffer/preamplificaton stage is needed before the regenerative stage. The preamplifier can also be designed to improve the speed of the comparator.

A simple implementation would be to simply place a static differential amplifier before the regenerative stage. This can simply be the following:

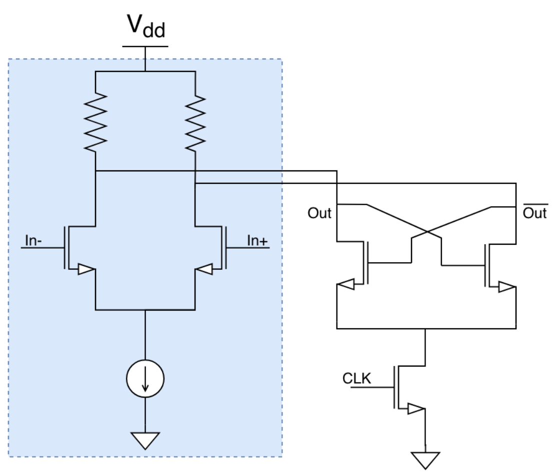


Figure 27: Resistive preamplifier, highlighted in blue, for buffering latch stage

Although this solves the main problem, it can be improved. The main disadvantage with this configuration is that the preamplifier draws static power. Additionally, the gain and bandwidth of a static preamplifier is not excellent. A better topology for a preamplifier is an integrating preamplifier. An integrating preamplifier further exploits the presence of a clock signal by obtaining gain via time.

The integrating amplifier uses on parasitic capacitances C_{par} for integration [20]. Initially, when the clock is low, the amplifier precharges the parasitic capacitances to V_{dd}. The PMOS transistors act as shorts, and the tail NMOS is open; the equivalent circuit is shown in 28 (c). When the clock is high, integration occurs. Assuming In- is smaller than In+, an unequal amount of current flows through the transistors and thus the parasitic capacitors (by KCL). As $\frac{I \cdot t}{C} = V$ (by $I = C\frac{dV}{dt}$), a small difference in currents exiting the capacitor will lead to increasingly diverging voltages at

[20]These capacitances need not be parasitic; they can even be tunable capacitors for external trimming.

the capacitor, illustrated by ΔV_1 and ΔV_2 in 28 (b). This also briefly illustrates the speed of the preamplifier: small C values and large I values (set by the size of the tail transistor) will increase speed. This is gain that can be used in a later stage.

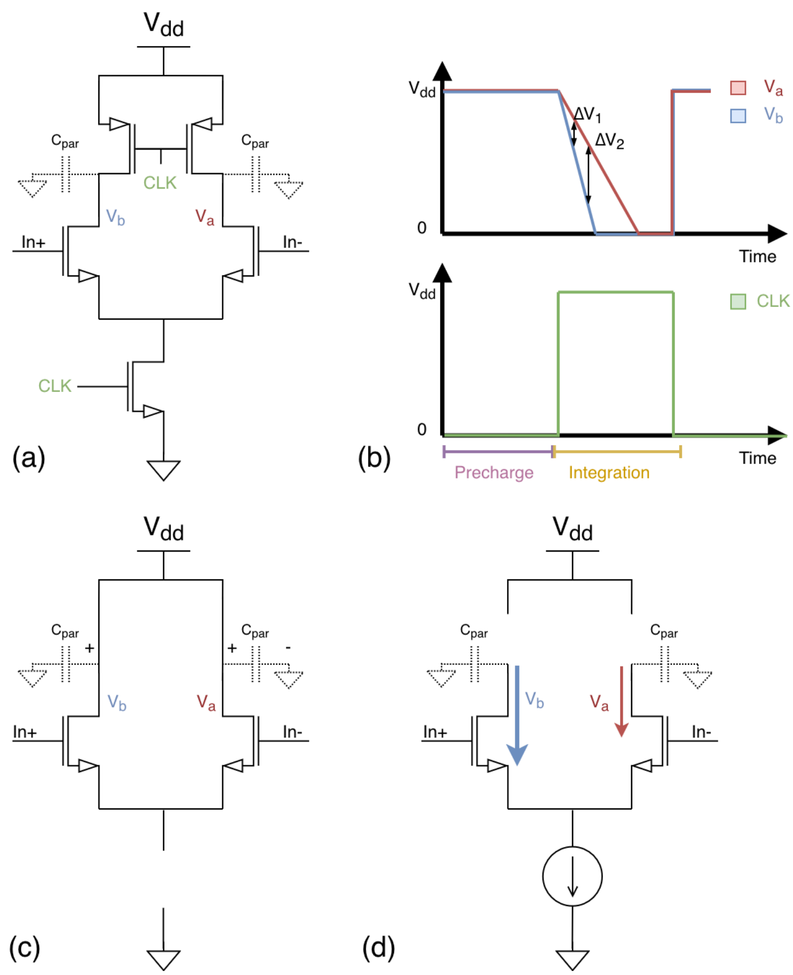


Figure 28: (a) Integrating preamplifier. V_a and V_b connect to the next stage. (b) Voltages during preamplifier operation asumming In- < In+ (c) Preamplifier in precharge state (d) Preamplifier during integration/amplification stage

4.c.3 Double Tail Comparator

The combination of these two stages (along with a few modifications) essentially forms the "double-tail comparator". After empirical testing with simulations, the double-tail comparator is a reasonable middle-ground of power consumption, speed, and kickback among StrongArm and triple-tail options. Resources on these comparators can be found here: [1, 26, 27].

This comparator topology is used in the FA24SP25 ADC. A starting design for the double-tail comparator appears as the following:

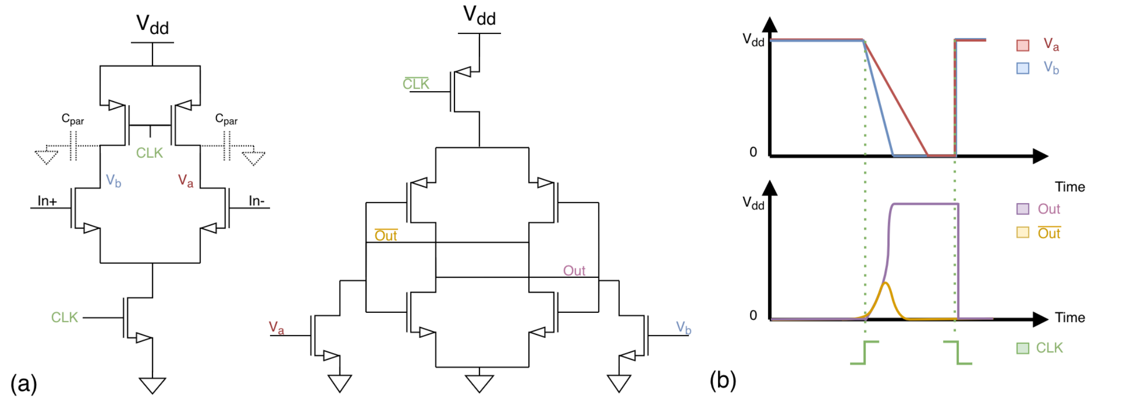


Figure 29: (a) Double-tail comparator (b) Waveforms of various nodes of double tail comparator.

In the double tail comparator, the preamplifier stage initially is precharged when CLK is low. The regeneration stage lies dormant as no current flows through it. At this moment, the output is invalid (both Out and $\overline{\text{Out}}$ are low) As soon as CLK goes high, the preamplifier begins integrating and the regenerative stage is activated as current now flows through it. When the regeneration is finished[21], the output is valid until the next precharge begins.

It is somewhat undesirable to have times where the output is valid and times when the output is not valid. It would be useful to implement logic that ”holds-over” the last known valid state whenever the output of the comparator is invalid (during precharge). This can be simply implemented using an SR latch [28].

4.c.4 Implementation

The FA24SP25 ADC has some important modifications to this design, namely the presence of a cascode structure and the use of low voltage threshold devices. The cascode structure (M4 and M5) in the preamplifier performs differential kickback isolation (which will be discussed in further detail) and reduce the capacitance of V_a and V_b. [29].

[21] This can be checked by Valid $=$ Out $\oplus$ $\overline{\text{Out}}$

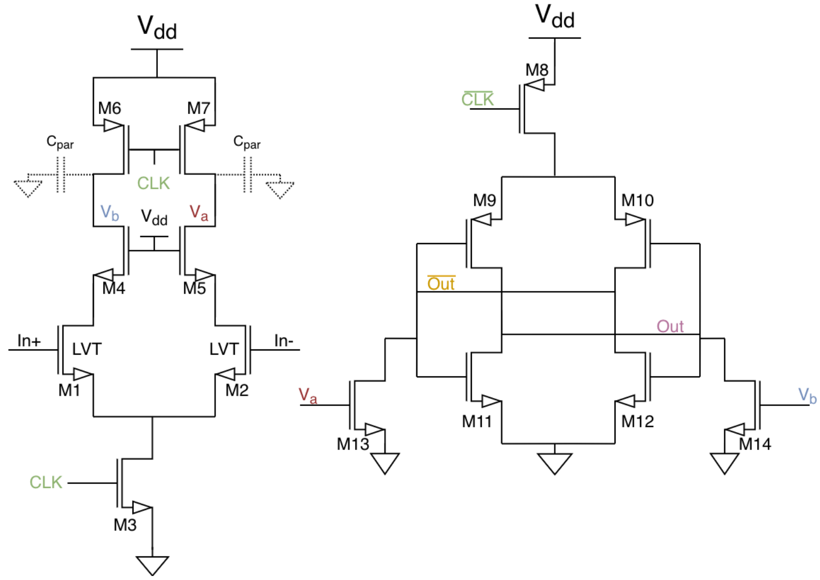


Figure 30: Modifications to the double-tail comparator in the FA24SP25 ADC.

Additionally, the input pair (M1 and M2) is made to be a lower voltage threshold device to improve the common mode input range of the comparator. In fully differential mode operation (and the input common mode limitation is obeyed), this should never be an issue, as the comparator common mode should always be $V_{DD}/2$. However, in truly differential mode, this may not necessarily be the case. If both In+ and In- becomes close to or lower than the threshold voltage of the input pair, both input pair transistors turn off [22]. This will greatly reduce the speed that C_{par} gets discharged and may cause the comparator to compare improperly.

The small signal common mode rejection ratio is not of a concern in this ADC, however. That is, the comparator does not need to handle changes in common mode voltage entering the comparator (asumming the common mode voltage does not exit the common mode range). This is because the IMCS switching algorithm causes both In+ and In- to revolve around the same common mode (and have the same common mode voltage) during the switching cycles.

It is important to briefly discuss transistor sizing in this comparator circuit. As common with the input pairs of most differential amplifiers, the input pair (transistors connected to In+ and In-) are the most significant contributors to noise and random offset. Random offset occurs due to manufacturing tolerances – M1 may have slightly different properties than M2, even if they are sized the same in schematic. If M1 is slightly wider than M2, for example, it will conduct more current even if In+ = In-. This can be modeled as a random *constant* DC voltage, V_{os} at the input of the comparator. Noise occurs due to random processes inherent to the physics of the transistors; random fluctuations due to heat contribute to noise. Noise is a *time varying* phenomenon modeled as voltage V_n^*. The non-ideal, noisy comparator can be modeled in Figure 31. Keep in mind that

[22]They are actually in subthreshold and will work if the ADC is clocked very slowly

this figure ignores some other non-idealities such as current/kickback noise.

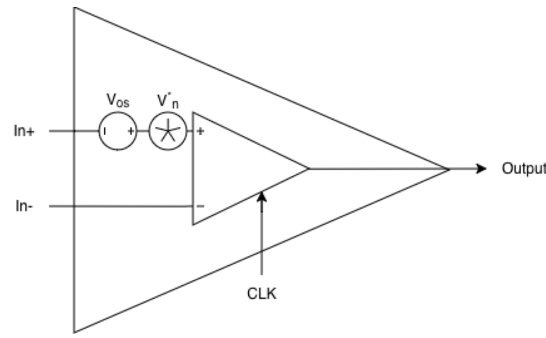


Figure 31: Comparator with noise and offset. The comparator inside is an ideal comparator.

To reduce the unavoidable effects of noise and offset, the input pair should be sized longer and wider. Noise and offset both decrease with increased transistor width and length. The mathematical description of offset can be described with the following relation (known as Pelgrom's law) [1]. Conceptually understanding noise is more involved and will be skipped here.

$$\sigma^2_{\Delta P} = \frac{A_P^2}{WL} \tag{11}$$

Conceptually, the variance of some transistor parameter due to *random* process variation is inversely proportional to width and length (area) of the transistor. A_P is some empirical proportionality constant. The standard deviation $\sqrt{\sigma^2_{\Delta P}}$ is thus inversely proportional to the square root of the area. An important distinction is to be made here – the process variation must be random – non-random variations between two devices (e.g. due to layout or thermal gradients) is systematic offset. Random offsets can arise from phenomena such as random dopant fluctuation (RDF) and materials variations.

To more systematically explore contributors of noise and offset, monte-carlo mismatch analysis was performed on all the transistors in the comparator. The length and width of M1 and M2 were increased such that. Additionally, the sensitivity analysis/mismatch contributors feature was used with Cadence Spectre, which corroborated our conceptual understanding of offset (the input pair dominated in offset). Other transistors important for offset include M13, M14, M9, and M10 [23]. Noise contribution analysis revealed the same transistors contributed the most noise.

Another note is that the tail transistors M3 and M8 govern the amount of current flowing through the comparator – smaller M3 and M8 transistors means slower "slewing" of the comparator, but also means less current consumption.

[23] M9 and M10 were somewhat surprising to the author. I hypothesize this is due to not enough integration/gain occurring before regeneration.

4.c.5 Kickback and Kickback Reduction

In dynamic comparators, a significant problem that occurs, which is insignificant in static comparators, is kickback. Kickback occurs when rapidly changing nodes, such as the source of M1, capacitively couples current back into the inputs. Kickback can either be common mode, meaning the quantity of coupling occurs equally between M1 and M2, or differential mode, meaning the difference in the amount of coupling seen at M1 and M2. Common mode kickback is not problematic as long as impedance seen at M1 equals that of M2 (i.e. the R_{eq} values are the same in Figure 32 (c)). It is still important to keep common mode kickback small if R_{eq} is different. Differential kickback is much more problematic as it may affect the comparison result.

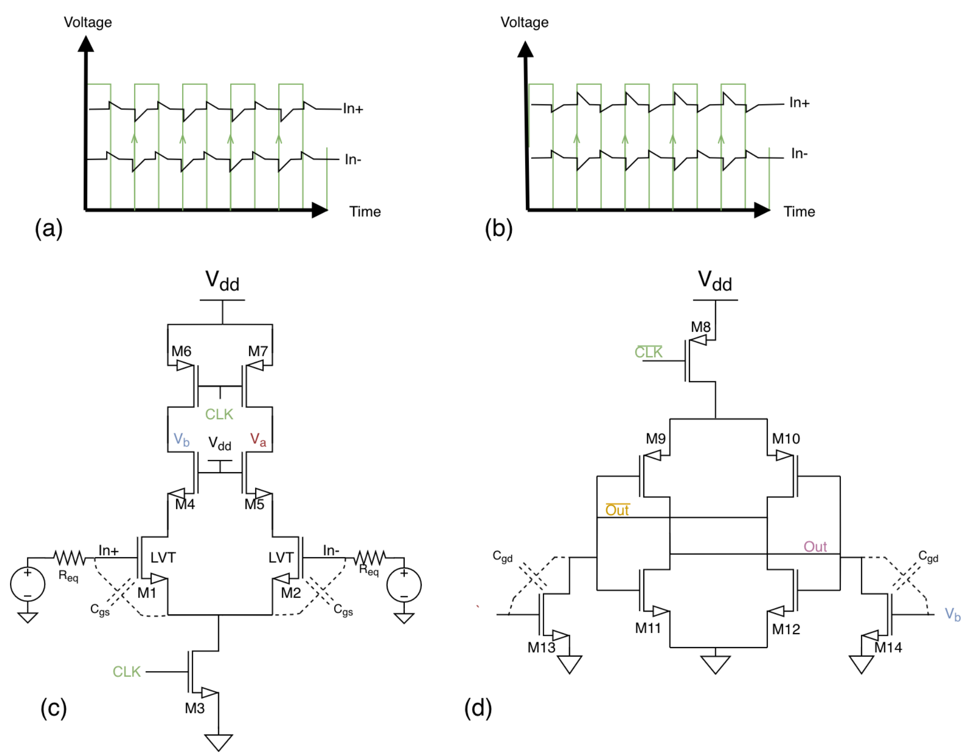


Figure 32: (a) Purely common mode kickback (b) Purely differential mode kickback (c) Source of common mode kickback (d) Source of differential mode kickback

One source of common mode kickback is shown in 32 (c). Changes in the source voltage of M1 and M2 are equal and can capacitvely couple current through C_{gs}. In the FA24SP25 tapeout, this was found to contribute the majority of the kickback.

A source of differential mode kickback is present in (d). As Out and $\overline{\text{Out}}$ slew in opposite directions during regeneration, the currents through C_{gd} are opposite, contributing to differential kickback. The double-tail architecture shown here makes this kickback insignificant because C_{gd} is usually small, and the cascode structure (M4 and M5) provides further isolation [26, 29].

One important point is that the load seen at In+ and In-, unlike what is shown in 32, is not resistive in an ADC with a CDAC. The CDAC is an array of capacitors and thus the load is nearly purely capacitive [26]. The issue posed by this is now all kickback is integrated by the capacitors. This means that if the kickback into the capacitors is different than the kickback out of the capacitors, this difference can collect over time. Thus, kickback into a capacitive source can be classified as peak kickback and integrated kickback.

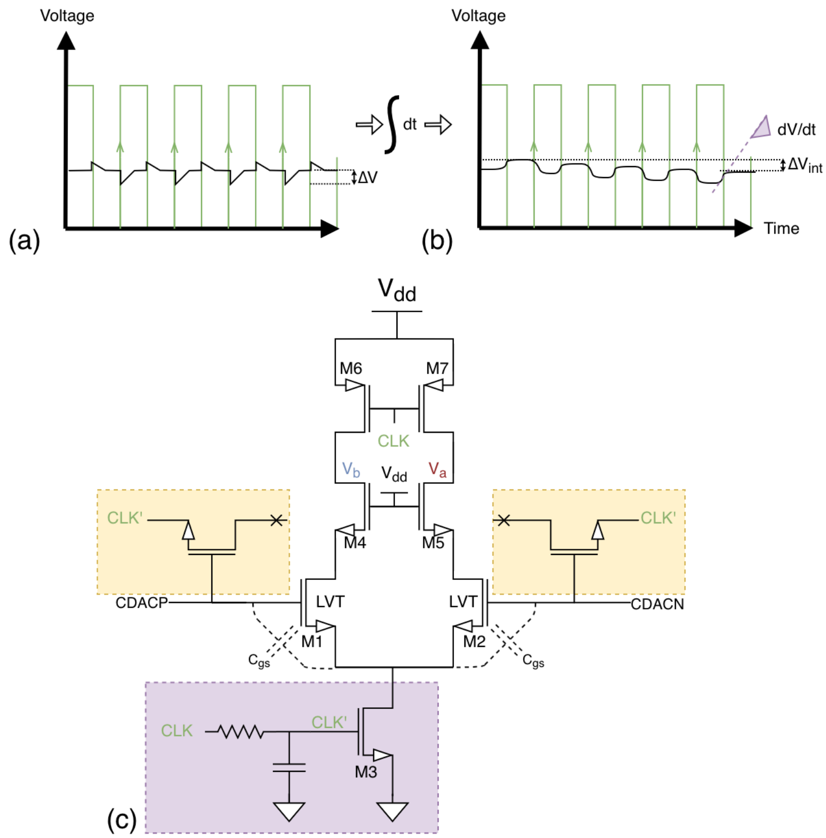


Figure 33: (a) Illustration of peak kickback in a circuit with resistive input sources (b) Illustration of integrated kickback in a circuit with capacitive input sources. Peak kickback manifests as a slope (shown in purple) (c) Preamplifier with kickback cancellation in FA24SP25 ADC. Slew-rate reduction (purple); kickback cancellation (yellow).

To reduce peak kickback, we can simply reduce the slew rate, or the rate that M3 turns on and off. This reduces the amount of current entering and leaving through C_{gs}. This adversely affects the speed of the comparator, however. As speed was not a concern in the ADC, we applied some slew rate reduction to the comparator.

To reduce integrated kickback, cancellation circuitry that creates current equal and opposite of that created by kickback can be used, similar to that of charge injection cancellation in analog switches. The gate source capacitance of an NMOS (of equal length and width of M1 and M2), has approximately the same C_{gs} as M1 and M2, performs this action [24]. As CLK (or CLK') increases, the voltage at the source of M1 and M2 decreases, current exits through $C_{gs,M1}$ and $C_{gs,M2}$. We cancel this by pushing current in through via the C_{gs} of the compensation transistors.

On a sidenote, CLK and $\overline{\text{CLK}}$ need to be generated. The generation circuitry to do so is shown below:

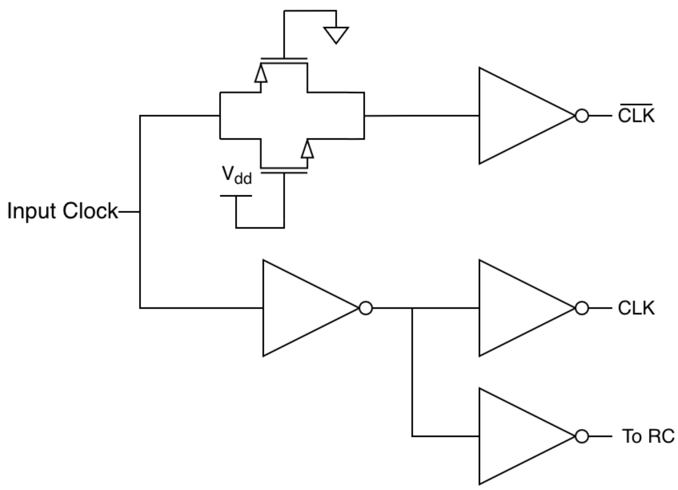


Figure 34: Clock generation circuitry

The always-on transmission gate is to impart a slight amount of extra delay to ensure that CLK and $\overline{\text{CLK}}$ are aligned in delay[25]. The "To RC" path has a smaller sized inverter to aid with slew rate limitation. This replaces the CLK that enters the slew rate reduction circuitry (shown in purple in 33 (c)).

4.d Digital Logic Implementation

The digital logic control circuitry plays a critical role in allowing the SAR algorithm to occur. A special quirk about the digital logic implemented in the FA24SP25 ADC is that the comparison occurs on the falling cycle of the master ADC clock, and the digital logic triggers on the rising edge of the master ADC clock. This was intentionally done to remove power consuming glitches from the CDAC logic.

The glitches occur because the previous switching cycle's comparison is needed before a decision for the next cycle is made. Since the comparator has a considerable delay, the digital logic is output is invalid while the comparator is making its decision (shown in shown in 35 (a)). The digital logic will output a bogus value, causing the CDAC drivers to randomly switch. Although the output will properly settle when the comparator finishes, these random switching actions waste power as

[24] They are not exactly the same since the channel in the MOSFET (governed by V_{DS}) is different
[25] This technique is not mine, it can be found on page 153 of [25]

driving the CDAC is a power hungry operation.

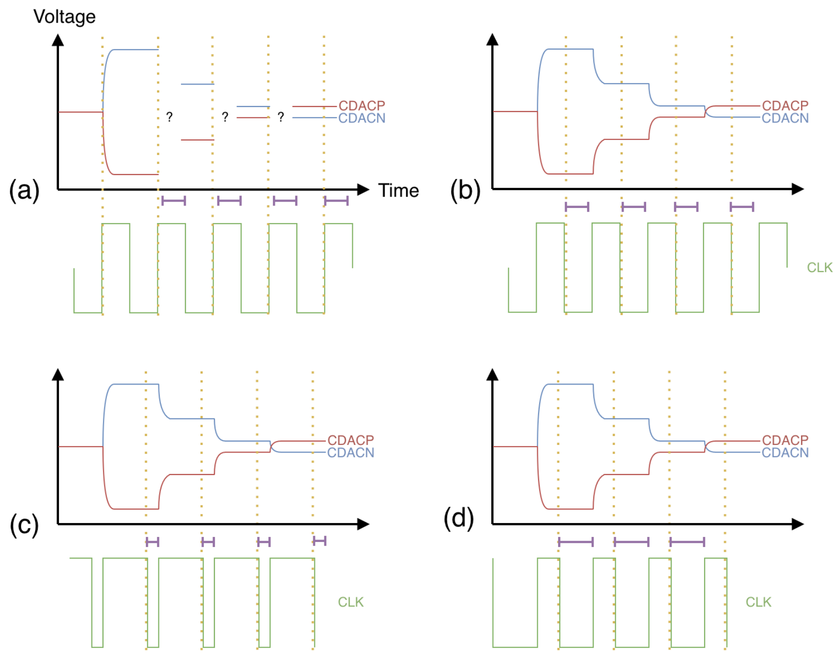


Figure 35: (a) Glitches due to timing (b) Solution using falling edge clock (c-d) Adjustability of clock duty cycle to account for various delays. Comparator delays are shown with purple lines

To mitigate this, we have the comparator sample on the falling edge of the clock so that it is done by the time the digital logic needs the comparison, shown in 35 (b). The drawback of this is less settling time is given to the CDAC; however, this can be ameliorated by adjusting the duty cycle of the input clock, if comparator delay is small enough. This is shown in Fig. 35 (c). Finally, if the comparator's delay is unexpectedly large, the duty cycle can still be adjusted to account for this in 35 (d).

4.e Layout Extra Considerations

Plenty of miscellaneous layout tips for reliability and reducing process variation can be found scattered across the internet, in various books, and sometimes in the PDK's Design Rule Manual (DRM) itself. There is a variety of resources and rules of thumb such as maintaining the same orientation between matched devices, not placing metal over transistor gates [30]... Many books also delve into electromigration (EM) rules and reliability considerations [31,32]. A primer to some important layout techniques are discussed below, but keep in mind that plenty of other considerations beyond what is discussed here exist.

4.e.1 Systematic Offset

The CDAC and comparator have special layout requirements. Noted in the discussion about the comparator, random offset is important to consider in the design and can be tested with Monte-Carlo simulation. However, layout has the ability to create *systematic* offset. These are mismatches within layout and are significant factors adding to the complexity of the CDAC and comparator layout.

Take for example the mock integrating preamplifier layout in Figure 36. The routing parasitic capacitances present at node V_a are not equal to those at node V_b due to differences in M1 routing, such that $C_{par1} \neq C_{par2}$. This increase in capacitance at node V_a will always occur across all devices that are fabricated – this is not a random process. With the understanding of how integrating preamplifiers work, this will create a *systematic* offset and not one that is random based on process variation.

Since these offsets are not random, they can be predicted and removed. Parasitic extraction is the tool used to extract layout/routing parasitics and backannotate them into the schematic, allowing the designer to simulate the schematic with layout imperfections. The systematic offset can then be corrected by modifying the layout.

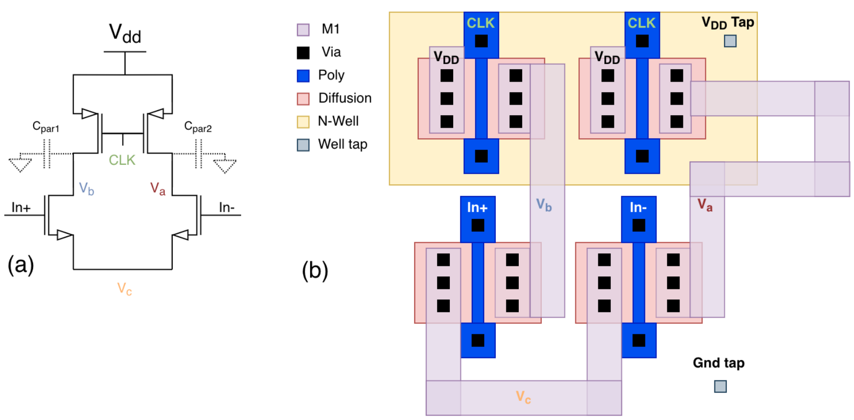


Figure 36: (a) Schematic of mock layout (b) Mock layout with systematic offset (increased capacitance on node V_a)

There are numerous other factors that can create these systematic offsets other than routing parasitics. For example, substrate/well tap placement. Here, one transistor is placed closer to a tap than the other transistor. Although the exact offsets caused by this are less predictable (potentially larger body effect in one transistor than the other), this can create offsets.

4.e.2 Process Gradients and Common Centroid

A factor that is more difficult to predict is process gradients and thermal gradients across the chip. These gradients will still create offsets. For example, a thermal gradient caused by hot spots on the chip may affect poorly designed circuitry. Additionally, chip manufacturing processes, such as ion implant and etching, can also create process variation gradients across the wafer and chip. To counteract the effect of these gradients, symmetry can be incorporated in the layout. Symmetry in the layout can be systematically added with common centroid layout [30,31].

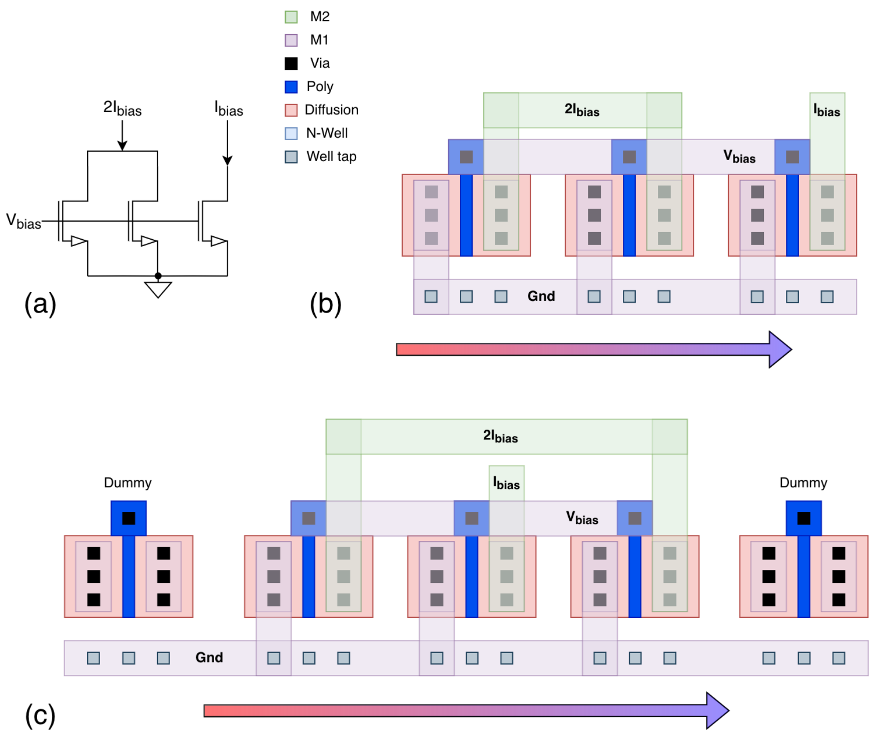


Figure 37: (a) Schematic of current mirror circuit (a) Layout of current mirror circuit sensitive to process/thermal gradients (b) improved current mirror layout

Furthermore, dummy devices may be added to the layout to reduce edge effects. In 37 (b), the leftmost and rightmost transistors experience different process effects as they are on the edge and have no surrounding transistors. To mitigate this, layout (c) adds dummy devices to pad the transistors and reduce these edge effects [26]

These process variations are also important to the CDAC, thus the CDAC was designed with common-centroid design in mind. As previously explored, a basic 7 bit binary weighted CDAC has $2^7 = 128$ capacitors. One of the two binary weighed CDACs (one being CDACP and the other being CDACN) in the FA24SP25 tapeout has 64 more unit capacitors for swing reduction, totaling to $128 + 64 = 192$ capacitors. The CDACP and CDACN were kept separate as combining them both into a single 384 capacitor common centroid configuration was deemed unnecessary and cumbersome. This large array of capacitors needs to be designed as symmetrically as possible to reduce

[26] These layout dummy devices are different to the "dummy" capacitor mentioned in the CDAC. They coincidentally are called the same in literature.

process variation due to gradients.

*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
*	I	I	I	I	I	I	I	I	I	I	I	I	I	I	*
*	I	I	I	A	B	A	B	B	A	B	A	I	I	I	*
*	I	I	I	B	A	B	A	A	B	A	B	I	I	I	*
*	I	A	B	A	A	C	A	A	C	A	A	B	A	I	*
*	I	B	A	A	A	D	A	A	D	A	A	A	B	I	*
*	I	A	B	C	D	E	C	C	E	D	C	B	A	I	*
*	I	B	A	A	A	C	G	F	C	A	A	A	B	I	*
*	I	B	A	A	A	C	F	H	C	A	A	A	B	I	*
*	I	A	B	C	D	E	C	C	E	D	C	B	A	I	*
*	I	B	A	A	A	D	A	A	D	A	A	A	B	I	*
*	I	A	B	A	A	C	A	A	C	A	A	B	A	I	*
*	I	I	I	B	A	B	A	A	B	A	B	I	I	I	*
*	I	I	I	A	B	A	B	B	A	B	A	I	I	I	*
*	I	I	I	I	I	I	I	I	I	I	I	I	I	I	*
*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*

Figure 38: CDAC capacitor arrangement. ”A”-”G” are capacitors weights from MSB to LSB, ”H” is the CDAC ”dummy” capacitor, ”I” are the swing reduction capacitors, and ”*” are the layout dummy capacitors to reduce edge effects.

A sidenote about the CDAC layout is that inter-capacitor edge effects occur, creating differences in parasitic capacitances. For example, here, ”A” capacitors will experience different parasitic capacitances compared to ”D” capacitors. This is because more ”A” capacitors are surrounded by other ”A” capacitors, whereas ”D” capacitors are not surrounded by any other ”D” capacitor. Thankfully, parasitic extraction can identify these differences and the CDAC layout can be adjusted to reduce them (and has already been done so to an acceptable value in this tapeout).

4.e.3 Latchup, Well Taps, and Substrate

This is a frequently overlooked/confusing consideration, but especially important for tapeout. The collection of design rule check (DRC) errors and LVS errors that crop up caused by the lack of understanding of this topic are quite confusing to troubleshoot. This topic is vital in creating well-designed layouts (and preventing the chip from exploding upon power up!).

Well taps/substrate connections are often a point of confusion. All transistors on chip are in fact 4 terminal devices: Drain, Gate, Source, Body/Bulk. The body connection of the transistor is often ignored in circuit analysis because it is tied to ground (for NMOS devices) or tied to V_{DD} (for PMOS devices). It is still important to keep this in mind because this connection has to be made in layout. Well and substrate taps allow the designer to make the 4th body connection in layout, shown in figure 39 (a).

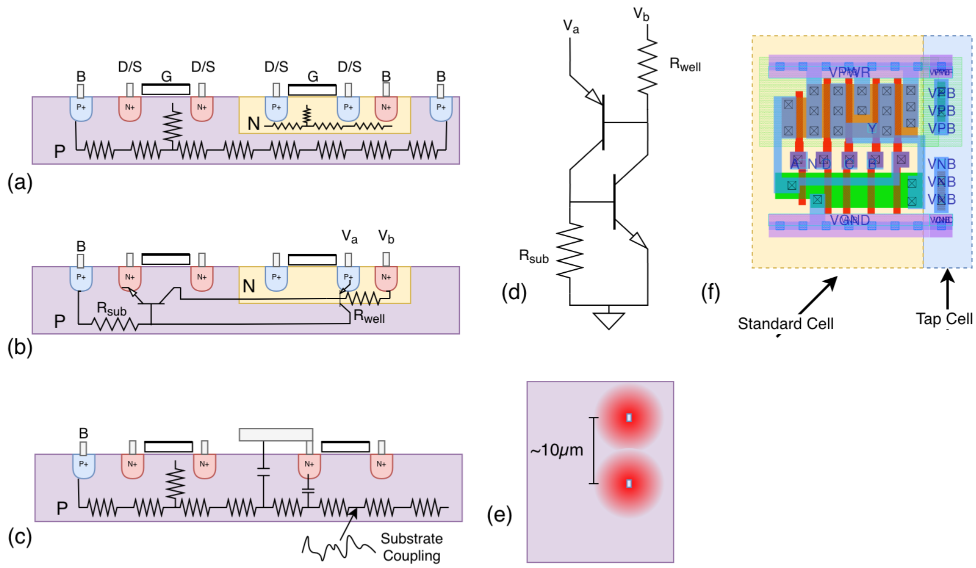


Figure 39: (a) Well/substrate taps acting as body connections. (b) Parasitic BJT devices formed in typical CMOS processes. (c) Substrate coupling affecting transistor operation. (d) Schematic of parasitic BJT devices in (b). (e) Seperation and operation of taps. (f) SKY130 Standard cell layout and tap cells.

Since the P-substrate and N-wells are conductive, a body connection does not need to be placed for every transistor. However, the substrate and wells are not excellent conductors, so wells need to be placed with a certain maximum spacing (defined by the PDK). Otherwise, substrate noise or, worse, latch-up can become problematic.

Figure 39 (b) depicts parasitic devices formed typical planar CMOS processes. These devices formed can cause severe problems such as latchup, but the formation of these devices are unavoidable. The schematic of the formed devices shown in 39 (d) outlines a latch-type circuit. When any of the parasitic bipolar transistors turn on, both will stay on and effectively short V_{DD} to Gnd until the power supply is removed or the chip destroys itself. This condition can be triggered by a variety of factors including substrate coupling, V_a rising above V_{DD}, ESD and more. Details are covered in depth in [31]. To reduce the likelihood of this happening, R_{sub} and R_{well} should be as small as possible to reduce the chance that the base voltages swing enough to activate these devices. This means that the maximum spacing of a device to a well tap or a substrate tap should be smaller than a few dozen microns [30] – $10\mu m$ should be safe for most applications [27].

The taps have a ”radius of action”, meaning they have a certain radius in which they can effectively tie down the substrate, depicted in 39 (e). As transistors move farther from the tap cell radially, the effectiveness of the tap cell decreases as the substrate resistances begin to accumulate. For precision analog applications, substrate coupling (shown in 39 (c)) from adjacent nodes may occur due to the resistance of the substrate. For sensitive devices, such as an input pair, it may be judicial to

[27] Some special ESD devices or devices that are ”injectors” may need guard rings/closer tap spacing [30, 31]

surround the entire transistor with well taps (in a guard ring configuration).

Finally, in standard cell layout (39 (f)), standard cells may not come with their own well taps or substrate taps. The standard cells need to be abutted with a tap cell every once in a few dozen microns (typically specified in PDK documentation) to prevent the aformentioned issues.

Below are some mistakes that can happen during schematic design and layout.

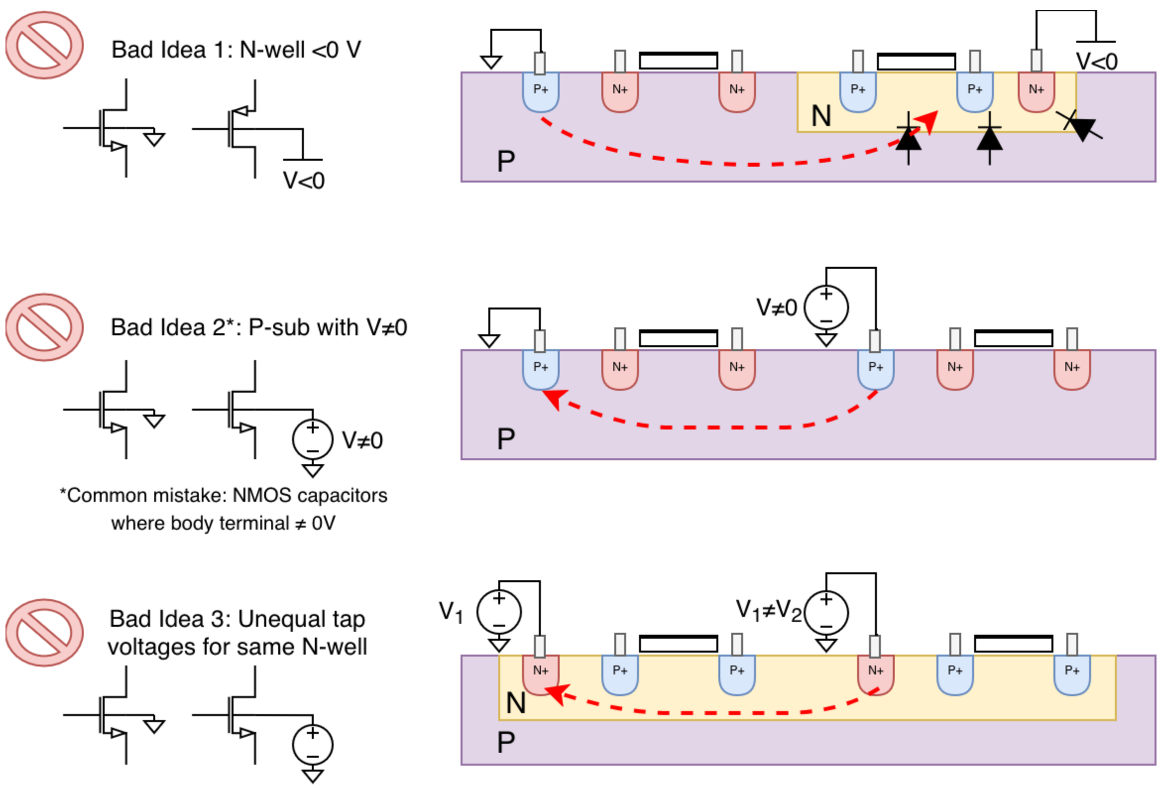


Figure 40: Bad ideas in schematic/layout design

Bad idea 1: it is a bad idea to connect an N-well to a voltage below 0V. The N well and the P substrate form a diode connection. Thus, it is a bad idea to connect the N-well (or the body of a PMOS) to a voltage that is lower than ground as it will cause forward biasing of this diode. They can be tied to any other voltage (within limits, of course). This should typically not be an issue unless PMOS MOSCAPs are used improperly.

Bad idea 2: it is a bad idea to connect the P-substrate to a nonzero voltage. One of the most important things to realize is that the entire p-substrate, and thus all NMOS transistor (except for Deep N-well) bodies are electrically tied to each other. The P substrate should be tied to ground at all times (with rare exceptions for back-biasing). This means it is not allowed to connect certain NMOS bodies to different voltages than ground. It is certainly not allowed to have various body voltages around the chip. This is a common mistake in using NMOS capacitors – the body terminal of NMOS capacitors must be connected to ground. Otherwise, they should be PMOS capacitors.

Bad idea 3: it is a bad idea to connect the same N-well to different voltages. Similarly to bad idea 2, when two transistors sit in the same well/substrate, their body are electrically tied together. Connecting the same well to different voltages will result in a short circuit. A simple fix is to place the two transistors in 2 different wells.

4.e.4 Antenna Rules

In modern CMOS processes, designers must take precautions when creating long stretches of metal – during the reactive ion etching process, charge can build up on these stretches and damage transistor gates or capacitors [31]. This poses special challenges when designing the CDAC. Additionally, there may be a separate DRC deck when checking for antenna rule violations. As these rule violations are quite tedious to fix, it is a good idea to be aware of these rules before layout.

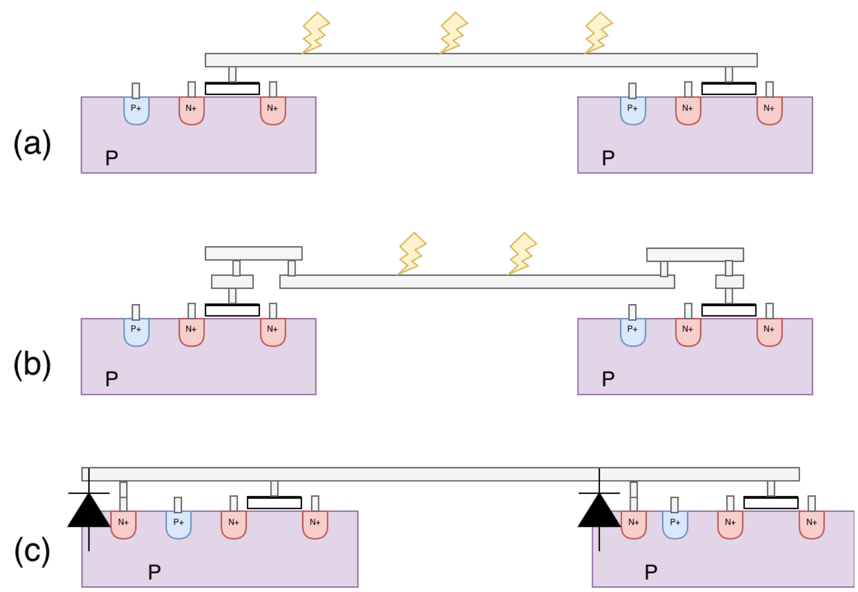


Figure 41: (a) Antenna rule violation due to long metal trace (b) Ascending metal layer method (c) Antenna diode method

There are a few simple methods in eliminating antenna effect and fixing antenna rules. One method is to ascend a layer of metal close to the gate, and immediately descend afterward. This fixes the antenna rule violation because the lower, longer, stretch of metal is manufactured first. When it is manufactured, it is not connected to the gate yet. Before the second layer is manufactured, the charge on this layer is dissipated and thus does not damage the transistor gates.

The second method is to add antenna diodes by adding connections to the P substrate through N+ diffusion. This forms a diode to the substrate (ground). This provides a small leakage path through a reverse biased diode during the etching process to dissipate away charge present on the long metal traces [28].

5 Acknowledgments

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- Annalise Thompson: Tapeout contributor
- Lena Conde Araujo: Tapeout contributor
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6 Appendix

6.a Input Common Mode Voltage Limitation

6.a.1 Derivation

To understand what gives rise to this limitation, not only does the flipping operation have to be considered, but also the switching algorithm in the first switching/comparison cycle (the cycle after flipping) has to be scrutinized.

As explored previously, during the first flipping stage (step 2-3 in Fig. 14, the voltage at the top plate of CDACP changes by $\Delta V_{topPflip} = V_{CM} - V_{IN+}$ and the top plate of CDACN changes by

[28] A slightly different method may be used in very modern processes [31]

$\Delta V_{topNflip} = V_{CM} - V_{IN-}$. There are two cases that the algorithm can take between step 3-4 (the up or down arrow, depending on the comparator output).

In the case that $V_{CM} + \Delta V_{topPflip} < V_{CM} + \Delta V_{topNflip}$, the comparator outputs a low value and the algorithm takes the up arrow in the second stage to increase the CDACP top plate voltage (in order to converge the two voltages). Similarly, the CDACN voltage is lowered to converge the two values. A diagram of this is shown in Fig. 14 (a), where the input to the ADC originally was $V_{IN+} = 0.95V$, $V_{IN-} = 0.05V$, $V_{CM} = 0.5V$. In this case, the voltages on the top plate of CDACP (red) and CDACN (blue) would remain between 0V and 1V. The opposite of this process would occur if $V_{CM} + \Delta V_{topPflip} > V_{CM} + \Delta V_{topNflip}$ – the CDACP voltage would be lowered and the CDACN voltage would be increased.

However, take for example if $V_{IN+} = 0.05V$, $V_{IN-} = 0.01V$, and $V_{CM} = 0.5V$. Notice that these voltages are both between the supply rails $V_{REF} = 1V$ and $GND = 0V$. Here, $\Delta V_{topP1} = V_{CM} - V_{IN+} = 0.45V$, $\Delta V_{topN1} = V_{CM} - V_{IN-} = 0.49V$. Then, $V_{CM} + \Delta V_{topPflip} < V_{CM} + \Delta V_{topNflip}$ (i.e. $0.95 > 0.99$). The comparator would output a low value and the algorithm takes the up arrow, which would cause the CDACP value to increase. There is a problem, however. The voltage on the top plate of CDACP is already at $0.95V$, any appreciable increase would exceed $V_{REF} = 1$! **Therefore, the point is, that there are further input voltage limitations that have to be imposed.**

To find these input voltage limitations, an equation for a general case of input voltages can be written. Here, we assume that $V_{REF} = 2V_{CM}$. Considering CDACP, the top plate of the CDACP is initially V_{CM}. After the first flip stage, the voltage at the top plate of CDACP is $V_{CM} + \Delta V_{topPflip} = V_{CM} + (V_{CM} - V_{IN+})$ Assuming the worst-case scenario where $V_{CM} + \Delta V_{topPflip} < V_{CM} + \Delta V_{topNflip}$, causing the CDACP's voltage to be further increased by ΔV_{topP1}, the voltage increase after the first comparison/switching stage (ΔV_{topP1}) can be expressed using Eq. 8.

$$\Delta V_{topP1} = \frac{C_1 \Delta V_1 + C_2 \Delta V_2 + ... + C_n \Delta V_n}{C_{tot}} = \frac{C_1 (V_{REF} - V_{CM})}{C_{tot}} = \frac{1}{2} V_{CM} \tag{12}$$

Thus, in the first switching stage, the voltage in worst case can increase by $0.5V_{CM}$ for CDACP and decrease by $0.5V_{CM}$ for CDACN.

If a designer wants to retain the entire input voltage range ($0 \leq V_{IN+}, V_{IN-} \leq V_{REF}$), another restriction must be imposed, or the architecture must be changed. One such restriction is that if CDACP is below V_{CM}, it should never become more negative. If CDACP is above V_{CM}, it should never become more positive. These restrictions can be applied similarly for CDACN. This restriction allows us to retain the whole input voltage range ($0 \leq V_{IN+}, V_{IN-} \leq V_{REF}$) because if any of the CDAC voltages is at 0V, it should never decrease below 0, and if any of the CDAC voltages is at V_{REF} it should never increase.

In practical terms, this means the following limits described in Eq. 9 [29]. Conceptually, this means that V_{IN+} should always be a mirror image of V_{IN-} across V_{CM} and thus in any case that $V_{IN+} > V_{CM}$, $V_{IN-} < V_{CM}$ and if $V_{IN+} < V_{CM}$ then $V_{IN-} > V_{CM}$. This means that the input voltage for CDACP can be expressed as $V_{IN+} = V_{CM} + \Delta V$ and$V_{IN-} = V_{CM} - \Delta V$ where ΔV is some input voltage.

Thus, the CDACP voltage equals $V_{CM} + (V_{CM} - V_{IN+}) = V_{CM} - \Delta V$ and the CDACN voltage also equals $V_{CM} + (V_{CM} - V_{IN+}) = V_{CM} - \Delta V$. Thus, if CDACP is above V_{CM}, CDACN is always below V_{CM}, and vice versa. This means that the comparator will always work towards converging both voltages to V_{CM} without causing the CDAC's plates to exit GND or V_{REF}.

6.a.2 Swing Reduction

Alternatively to imposing common mode input voltage limitations, the CDAC topology may be slightly modified to allow for arbitrary inputs. Previously, it was shown that the "worst case" scenario occurs when CDACP/N is already at a voltage near V_{REF} and the algorithm continues to try and increase CDACP/N, or CDACP/N is already at a voltage near GND and the algorithm continues to try and decrease CDACP/N. The previous restriction prevented this from happening in the first place.

Another way to alleviate CDACP/N from exceeding V_{REF} or GND is if somehow extra headroom is added, or the voltage changes are reduced such that even if the "worst case" is taken, the voltage will still remain in the limit. This can be done by adding capacitors that scale down the changes in voltage of the top plate at each switching operation [13]. Previously we seen that maximally, the voltage after the flipping operation (given $0 \leq V_{IN+}, V_{IN-} \leq V_{REF}$, is also $(0 \leq V_{CDACP}, V_{CDACN} \leq V_{REF})$ because $\Delta V_{topflip}$ (the change in voltage on CDACN or CDACP after the flip) is maximally V_{CM}. Additionally, the maximal ΔV_{top1} (the change in voltage on CDACN or CDACP after the first switching cycle) is $\frac{1}{2}V_{CM}$.

Arbitrary inputs can be accepted if there is a way to scale the change in voltage down such that the sum of the worst case scenarios (the sum of the worst case $\Delta V_{topflip} + \Delta V_{top1}$) never exceeds V_{CM}. This indeed can be the case if we scale down $\Delta V_{topflip}$ and ΔV_{top1} by 2/3 so that the worst case scenarios sum up to less than V_{CM} because $2V_{CM}/3 + (2/3) \cdot V_{CM}/2 \leq V_{CM}$. This can simply be achieved by adding an extra capacitor to the CDACP and CDACN tied to an AC ground of value equal to the MSB. A capacitor with size equal to the MSB capacitor tied to AC ground will equal $C_{tot}/2$ and will be multiplied by $\Delta V = 0$, and thus mean that the denominator of Eq. 8 will increase by $C_{tot}/2$ and become $1.5C_{tot}$, meaning that ΔV_{top} will be multiplied by 2/3.

[29] the limit can actually be more generous than this if you impose limits on the input voltage range, too. These limits are complicated for a designer to think about and a end-user to use, though.

6.b Analog Switch

6.b.1 Graph Derivations

To derive the on resistance of an analog switch (its associated R_{on}), one must make some assumptions of the transistors' operating regime in the analog switch. One can make an assumption that the analog switches are operating in deep triode region (most valid when the voltage across the analog switch is small). Another set of assumptions that can be made (and works fairly well) is ignoring second order effects such as body effect and assuming square law operation. Thus, the set of "textbook" standard analysis equations can be used. Namely, the following equation for NFETs [21]:

$$R_{dson} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})} \tag{13}$$

To find V_{GS}, the gate voltage of the on and off state must be defined (the switch input is a digital signal), which we can denote as V_{HIGH} and V_{LOW}. When the switch is fully on, the gate voltage of the NMOS is thus V_{HIGH} Additionally, the input voltages (IO1 and IO2 from Figure. 18) have to be defined. Since we already assumed the transistor is in deep triode, we must choose a realistic condition that fits this scenario, so we assume $V_{IO1} = V_{IO2}$. From here, we can construct the equations for the on resistance of the NFET ($R_{dson,n}$) and the on resistance of the PFET $R_{dson,p}$. Here V_{TN} and V_{TP} are the threshold voltages for NFETs and PFETs, respectively.

$$R_{dson,n} = \begin{cases} \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GSN} - V_{TN})} & V_{IO} < V_{HIGH} - V_{TN} \\ 10^{10} & \text{else} \end{cases}$$

$$R_{dson,p} = \begin{cases} \frac{1}{\mu_p C_{ox} \frac{W}{L}(V_{GSP} - V_{TP})} & V_{IO} > V_{LOW} - V_{TP} \\ 10^{10} & \text{else} \end{cases}$$

In the cases that the switch is on, the gate of the NMOS is at V_{HIGH} and the gate of the PMOS is at V_{LOW}. When the switch is on and $V_{IO1} = V_{IO2} < V_{HIGH} - V_{TN}$, then the NMOS is outside the cutoff regime and therefore has a appreciable conductance. When the switch is on, if $V_{IO1} = V_{IO2} > V_{LOW} - V_{TP}$ (noting that V_{TP} is typically negative), then the PMOS is outside the cutoff region and is conducting. If both these conditions are met, then both the NMOS and PMOS are cooperating. If the transistors are indeed in cutoff, the resistance is large and (arbitrarily) set to $10^{10}\Omega$. The resistance of the analog switch is then the parallel of the two ($R_{dson,n}||R_{dson,p}$)

Since the gate of the NMOS is at V_{HIGH}, $V_{GSN} = V_{HIGH} - V_{IO}$, and for the PMOS, $V_{GSN} = V_{LOW} - V_{IO}$. Sweeping V_{IO} (assuming $V_{HIGH} = 1.8V$, $V_{LOW} = 0V$ and $V_{TN} = 0.3V$ yields the following curve.

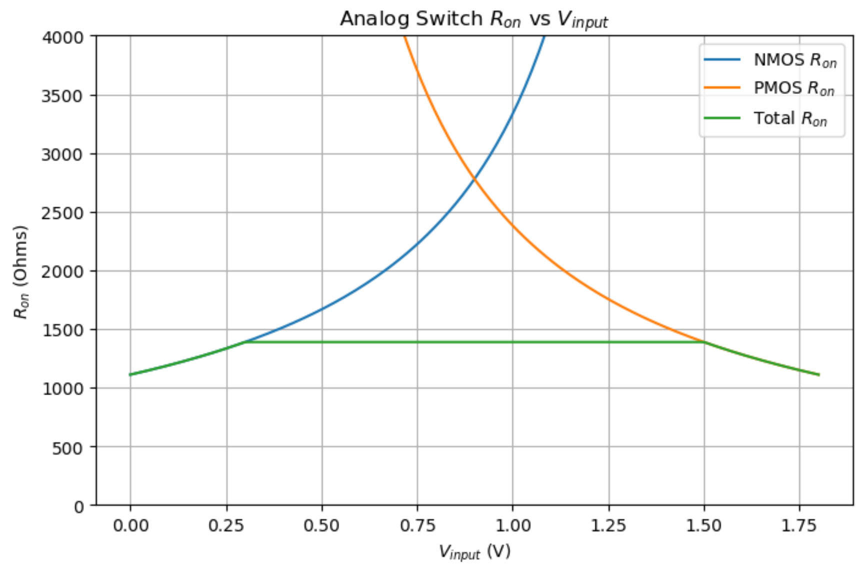

Figure 42: Initial model of analog switch resistance

Here, the analog switch on resistance appears correct, except for one major discrepancy. The resistance curve in the middle (when the NMOS and PMOS are both on), is flat. To combat this, an extra resistance term has to be added to the R_{dson} term [33]. Thus, the resistance of the switch has to be modeled as $(R_{dson,n} + R_n)||(R_{dson,p} + R_p)$ to generate the characteristic curve for an analog switch as seen in 19.

6.b.2 Bootstrapped Switch

In a bootstrapped switch, V_{GS} no longer changes due to V_{IO}, so the on resistance, at first order, is flat. This does not completely agree with simulation results as one assumption becomes necessary to consider: body effect.

TODO

6.c Other Comparators

TODO

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